

TriQ-KEM

Key Encapsulation Mechanism Specification and Supporting Documentation

Submitters: Lan Luo, Chaofeng Guan, Yunlei Zhao, Hong Wang, Jianhua Hou, and Tong Yu.

Contact: Lan Luo, lorraine0416@163.com

Abstract

TriQ-KEM is a code-based IND-CCA2 secure key-encapsulation mechanism obtained by applying the Fujisaki–Okamoto (FO) transform to TriQ-PKE, an IND-CPA secure public-key encryption scheme. The concrete hardness estimates rely on constrained-weight quasi-cyclic syndrome decoding (CW-QCSD) assumptions.

Compared with HQC, TriQ-KEM uses attack-aware decryption-failure evaluation, unbalanced weights for the PKE encryption vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}$, and public bounded-density rejection sampling. The submitted instances aim for a better balance among bandwidth/efficiency, decryption failure rate (DFR), and concrete security.

This specification defines the TriQ-PKE and TriQ-KEM algorithms and the submitted parameter sets, and reports implementation and performance results. It also gives correctness and decryption-failure bounds, density-interval cumulative decryption-failure (CDFR) evaluation, and security analysis.

Index Terms Code-Based Cryptography, Hamming Quasi-Cyclic, Decryption Failure, Information Set Decoding, Unbalanced Errors, Rejection Sampling.

CONTENTS

I	Introduction	3
I-A	Design Rationale	3
I-B	Organization	3
II	Preliminaries	4
II-A	Notations	4
II-B	Algebraic and Coding Conventions	4
III	Specifications	6
III-A	TriQ-PKE Algorithms	7
III-B	Bounded-Density Rejection Rule and Vector Sampling	9
III-C	TriQ-KEM Algorithms	9
IV	Parameter Sets and Sizes	10
IV-A	Parameter Selection	10
IV-B	Parameter Sets	11
IV-C	Key and Ciphertext Sizes	12
V	Security Analysis	12
V-A	Security Statement	13
V-B	Decryption Failure Rate Analysis	13
V-C	CW-QCSD and ISD Estimates	20
VI	Implementation and Performance	22
VI-A	Implementation Overview	22
VI-B	Performance Evaluation	23
VI-C	Known-Answer Tests	24
VI-D	Feature Statement	24
VI-E	Intellectual Property Notice	25
A	Proofs for Failure Bounds	26
A-A	BSC and Bounded-Density Rejection Proofs	26
A-B	Proofs for RM and RSRM Failure Bounds	27
A-C	Proofs for BSC and Weight-Redistribution Results	34
B	Structured-ISD Analysis for Unbalanced QCSD	37
C	HQC Parameters and Estimates for Comparison	39
C-A	HQC Parameter Sets Used for Comparison	39
C-B	HQC ISD Estimates for Comparison	40
C-C	HQC DFR and CDFR Estimates for Comparison	40
	References	41

I Introduction

I-A Design Rationale

The name TriQ combines “Tri” and “Q”: “Tri” refers to the three-way design balance among bandwidth/efficiency, decryption-failure rate (DFR), and concrete security, while “Q” refers to the quasi-cyclic code-based structure. TriQ-KEM is an IND-CCA2 secure key-encapsulation mechanism obtained by applying the Fujisaki–Okamoto (FO) transform to TriQ-PKE, an IND-CPA secure public-key encryption scheme. The concrete hardness estimates rely on the 2-CW-QCSD problem associated with public-key recovery and the bounded-density-conditioned 3-CW-QCSD problem associated with ciphertext-randomness recovery.

TriQ-KEM is characterized by design choices at the PKE level and an attack-aware decryption-failure evaluation criterion: an attack-aware CDFR criterion, a public bounded-density rejection rule, and unbalanced weights for the PKE encryption vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}$.

1. **Attack-aware CDFR criterion.** HQC parameter selection requires the natural DFR to stay below the target level, following its refined decryption-failure analysis [1]. The density-interval attack [2] shows that adversarially chosen ciphertexts can amplify the relevant failure probability. TriQ-KEM therefore evaluates the cumulative decryption-failure probability (CDFR) over the feasible density-interval attack domain and requires it to remain below the stated target. To limit the attack domain used in this evaluation, TriQ-KEM applies the public bounded-density rejection rule described below. The submitted parameter sets use the post-rejection CDFR bounds reported in Table 8.
2. **Public bounded-density rejection rule.** The sampler draws fixed-weight candidates for $\mathbf{r}_1$ and $\mathbf{r}_2$ and applies the public bounded-density rejection rule. Conditioned on acceptance before the attempt limit, the output is distributed over the accepted set defined in Section III-B. The rule is public, depends only on the support of each candidate fixed-weight sample and public parameters, and is independent of secret data. It rejects candidate fixed-weight samples for $\mathbf{r}_1$ and $\mathbf{r}_2$ containing more than γ nonzero positions in any cyclic window of length L_{bd} . This restricts dense cyclic intervals in the accepted sampling space. The rejection overhead is bounded by the probabilities in Table 4. The resulting decryption-failure bounds are evaluated in Section V, while the effect of the constrained sampling space on the 3-CW-QCSD estimate is accounted for in Section V-C.
3. **Unbalanced encryption-vector weights.** In HQC, the encryption randomness vectors and the error vector are parameterized with the same weight, $w_{\mathbf{r}} = w_{\mathbf{e}}$. TriQ-KEM instead uses a constrained-weight choice:

$$w_{\mathbf{r}} < w_{\mathbf{e}}, \quad \mathbf{r}_1, \mathbf{r}_2 \in \mathcal{R}_{w_{\mathbf{r}}}, \quad \mathbf{e} \in \mathcal{R}_{w_{\mathbf{e}}}.$$

The secret-vector weights are chosen as $w_{\mathbf{x}} = w_{\mathbf{y}} = w_{\mathbf{r}}$. This reduces the contribution of the product terms $\mathbf{x}\mathbf{r}_2$ and $\mathbf{r}_1\mathbf{y}$ to the decryption noise, improving the tradeoff among bandwidth/efficiency, decryption failure rate (DFR), and concrete security. In the security evaluation, public-key recovery corresponds to a 2-CW-QCSD problem for the secret pair $(\mathbf{x}, \mathbf{y})$, while ciphertext-randomness recovery corresponds to a bounded-density-conditioned 3-CW-QCSD problem for $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$, where the bounded-density condition applies to $\mathbf{r}_1$ and $\mathbf{r}_2$. The CW-QCSD formulation generalizes the standard QCSD setting used for HQC: the equal-weight case recovers QCSD, while TriQ-KEM uses prescribed per-block weights. The corresponding classical ISD and quantum-Prange bit-complexity estimates are reported in Section V-C.

I-B Organization

The rest of the document is organized as follows:

- Section II introduces notation, algebraic and coding conventions, and the SD/CW-QCSD definitions;
- Section III specifies the PKE code structure, bounded-density sampler, and PKE/KEM algorithms;
- Section IV lists the concrete parameter sets, RSRM instances, bounded-density rejection parameters, and KEM key, ciphertext, and shared-secret sizes;
- Section V gives the security statement, DFR and CDFR analysis, classical ISD estimates for the relevant CW-QCSD problems, quantum-Prange bit-complexity estimates, and the FO-transform IND-CCA2 security statement;
- Section VI gives the implementation overview, benchmark methodology and results, KAT format, feature statement, and intellectual-property notice.

The appendices contain supporting proofs, structured-ISD details, and the HQC parameter sets, cycle counts, and security estimates used for comparison.

II Preliminaries

This section fixes notation, algebraic and coding conventions, and the syndrome-decoding problem definitions used later. Vectors and polynomials are denoted by lowercase bold symbols, while matrices are denoted by uppercase bold symbols.

II-A Notations

The notation used in the algorithm description and security analysis is collected here. Table 1 lists the symbols used throughout the specification.

II-B Algebraic and Coding Conventions

Polynomial ring. Length- n binary cyclic vectors can be represented as elements of the quotient ring

$$\mathcal{R} = \mathbb{F}_2[X]/(X^n - 1),$$

where n is taken to be a primitive prime throughout this specification. A primitive prime makes the nontrivial cyclotomic factor of $X^n - 1$ irreducible over $\mathbb{F}_2$; this avoids splitting the quasi-cyclic representation into smaller irreducible components. An element $\mathbf{a} \in \mathcal{R}$ can be represented by its coefficient vector

$$\mathbf{a} = (a_0, a_1, \dots, a_{n-1}) \in \mathbb{F}_2^n,$$

corresponding to the polynomial $a_0 + a_1X + \dots + a_{n-1}X^{n-1}$. Addition in $\mathcal{R}$ is component-wise addition over $\mathbb{F}_2$. Multiplication of two elements $\mathbf{a}, \mathbf{b} \in \mathcal{R}$ is polynomial multiplication modulo $X^n - 1$:

$$\mathbf{ab} = \sum_{k=0}^{n-1} \left(\sum_{i+j \equiv k \pmod{n}} a_i b_j \right) X^k.$$

Equivalently, the coefficient vector of the product is the length- n cyclic convolution of $\mathbf{a}$ and $\mathbf{b}$.

Weight. Using the coefficient-vector representation above, define the Hamming weight of $\mathbf{a} \in \mathcal{R}$ as

$$\text{wt}(\mathbf{a}) = \#\{i \mid a_i \neq 0\}.$$

The support of $\mathbf{a}$ is $\text{Supp}(\mathbf{a}) = \{i \mid a_i \neq 0\}$. The Hamming distance between $\mathbf{a}, \mathbf{b} \in \mathcal{R}$ is $\text{wt}(\mathbf{a} + \mathbf{b})$. For a nonnegative integer w , $\mathcal{R}_w$ denotes the subset of $\mathcal{R}$ consisting of elements of Hamming weight w .

Table 1: Core notation used throughout the TriQ-KEM specification

Symbol	Meaning
$\mathcal{R}$	The ring $\mathbb{F}_2[X]/(X^n - 1)$ used by TriQ-KEM
n	Ambient ring length used by the cyclic polynomial arithmetic
n_1, n_2	Outer Reed–Solomon length and inner repeated Reed–Muller length in the RSRM construction
$n_1 n_2$	RSRM codeword length
k	RSRM code dimension and plaintext length in bits
ℓ	Difference $\ell = n - n_1 n_2$ between the ring length and the RSRM codeword length
C	Reed–Solomon/Reed–Muller (RSRM) concatenated code used for message encoding and decoding
$\mathcal{R}_w$	Binary polynomials in $\mathcal{R}$ with Hamming weight w
$A_{L,\gamma}^{(w)}$	Bounded-density accepted set of weight- w ring elements with at most γ nonzero positions in any cyclic window of length L
L_{bd}, γ	Cyclic-window length and upper bound on nonzero positions in each such window
N_{max}	Maximum number of bounded-density sampling attempts
U_{clump}	Upper bound on the bounded-density rejection probability
$\lambda_{\text{bd}}^{(w)}$	Entropy-loss term associated with bounded-density rejection at weight w
$w_{\mathbf{x}}, w_{\mathbf{y}}$	Secret-vector weights used for $\mathbf{x}$ and $\mathbf{y}$
$w_{\mathbf{r}}$	Common fixed weight used for the PKE encryption vectors $\mathbf{r}_1$ and $\mathbf{r}_2$; the specified instances satisfy $w_{\mathbf{x}} = w_{\mathbf{y}} = w_{\mathbf{r}}$
$w_{\mathbf{e}}$	Fixed weight used for the PKE encryption error vector $\mathbf{e}$
$ek^{\text{PKE}}, dk^{\text{PKE}}, c^{\text{PKE}}$	TriQ-PKE encryption key, decryption key, and ciphertext
$ek^{\text{KEM}}, dk^{\text{KEM}}, c^{\text{KEM}}$	TriQ-KEM encapsulation key, decapsulation key, and ciphertext
H, G, J	Hash and derivation functions used in the FO transform for TriQ-KEM
σ	KEM implicit-rejection secret used by decapsulation
θ	PKE encryption randomness output by G during KEM encapsulation
salt	Public salt included in each KEM ciphertext
$\mathbf{e}'$	Decryption noise term $\mathbf{x}\mathbf{r}_2 + \mathbf{r}_1\mathbf{y} + \mathbf{e}$
DFR	Decryption-failure rate for the TriQ-PKE decryption-noise distribution. In the density-interval analysis, $\text{DFR}(l_1, l_0)$ denotes the value when weak-ciphertext parameters (l_1, l_0) condition this distribution; natural DFR denotes the value under honest TriQ-PKE encryption
CDFR	Cumulative decryption-failure probability over the feasible density-interval attack domain under the specified bounded-query setting

Linear codes and quasi-cyclic codes. A linear code $\mathcal{C}$ of length n and dimension k over $\mathbb{F}_2$ is a k -dimensional subspace of $\mathbb{F}_2^n$; it is denoted by $[n, k]$. A generator matrix $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ satisfies $\mathcal{C} = \{\mathbf{u}\mathbf{G} \mid \mathbf{u} \in \mathbb{F}_2^k\}$. A parity-check matrix $\mathbf{H} \in \mathbb{F}_2^{(n-k) \times n}$ satisfies $\mathcal{C} = \{\mathbf{x} \in \mathbb{F}_2^n \mid \mathbf{H}\mathbf{x}^\top = \mathbf{0}\}$. The *syndrome* of a vector $\mathbf{x} \in \mathbb{F}_2^n$ with respect to $\mathbf{H}$ is $\mathbf{H}\mathbf{x}^\top$. Over $\mathbb{F}_2$, a linear code $\mathcal{C} \subseteq \mathbb{F}_2^n$ is *cyclic* if a cyclic shift of any codeword is also a codeword. As noted above, multiplication by X in $\mathcal{R}$ corresponds to a cyclic shift of the coefficient vector; hence the ideals of $\mathcal{R}$ are exactly the cyclic codes of length n over $\mathbb{F}_2$.

A code is *quasi-cyclic* of index s if applying the same cyclic shift to each of its s length- n blocks leaves a codeword. In the rate- $1/s$ case used below, such a code has length sn and dimension n .

For a systematic QC code of index s and rate $1/s$, the parity-check matrix has the form

$$\mathbf{H} = \left(\begin{array}{cccc|c} \mathbf{I}_n & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{H}_0 \\ \mathbf{0} & \mathbf{I}_n & \cdots & \mathbf{0} & \mathbf{H}_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_n & \mathbf{H}_{s-2} \end{array} \right),$$

where $\mathbf{I}_n$ is the identity matrix of size n and each $\mathbf{H}_i$ is an $n \times n$ circulant matrix. In the basic QCSD formulation, all blocks of the error vector share the same weight w . A natural generalization allows distinct weight constraints per block. This gives the constrained-weight s -CW-QCSD problem below. Let $\mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_s) \in \mathbb{F}_2^{sn}$ denote an error vector whose i -th block $\mathbf{e}_i$ has prescribed weight w_i .

Definition II.1 (s -CW-QCSD Problem). Let $s, w_1, \dots, w_s$ be positive integers. Given the parity-check matrix $\mathbf{H} \in \mathbb{F}_2^{(sn-n) \times sn}$ of a systematic QC code $\mathcal{C}_{\text{QC}}$ with index s and rate $1/s$, a syndrome $\mathbf{s} \in \mathbb{F}_2^{sn-n}$, and weight constraints $(w_1, \dots, w_s)$, find an error vector $\mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_s) \in \mathbb{F}_2^{sn}$, such that $\text{wt}(\mathbf{e}_i) = w_i$ for each $i = 1, \dots, s$, and $\mathbf{H}\mathbf{e}^\top = \mathbf{s}^\top$. This problem is denoted as $s\text{-CW-QCSD}(sn, w_1, \dots, w_s)$.

The s -CW-QCSD problem generalizes the standard QCSD problem by allowing per-block weight constraints; the special case $w_1 = \dots = w_s$ recovers the usual QCSD formulation.

Remark II.2 (TriQ-KEM instances). For TriQ-KEM, the concrete hardness estimates use the 2-CW-QCSD problem for public-key recovery with target $(\mathbf{x}, \mathbf{y})$, and the 3-CW-QCSD problem conditioned on bounded-density rejection for ciphertext-randomness recovery with target $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$. The bounded-density condition applies only to $\mathbf{r}_1$ and $\mathbf{r}_2$. Its reduction of the accepted sampling space is accounted for as an entropy-loss correction in the ISD estimates of Section V-C.

III Specifications

This section specifies TriQ-PKE and the TriQ-KEM construction. The distinguishing TriQ choices are made in the PKE component: unbalanced encryption-vector weights and public bounded-density sampling for $\mathbf{r}_1, \mathbf{r}_2$. TriQ-KEM is obtained by applying the Fujisaki-Okamoto (FO) transform to TriQ-PKE, yielding an IND-CCA2 secure key-encapsulation mechanism. In this section, $|salt|$ denotes the salt length in bytes; its submitted value is listed in Section IV. The section first gives the TriQ-PKE structure and algorithms, then specifies the bounded-density rejection rule, and finally gives the TriQ-KEM algorithms.

TriQ-PKE data flow

$$\begin{aligned}
\text{KeyGen}(seed^{\text{PKE}}) \rightarrow (ek^{\text{PKE}}, dk^{\text{PKE}}) : & \text{ derive } (seed_{\text{dk}}^{\text{PKE}}, seed_{\text{ek}}^{\text{PKE}}) \text{ from } seed^{\text{PKE}}, \\
& \mathbf{x} \xleftarrow{\$} \mathcal{R}_{w_x}, \quad \mathbf{y} \xleftarrow{\$} \mathcal{R}_{w_y}, \quad \mathbf{h} \xleftarrow{\$} \mathcal{R}, \quad \mathbf{s} = \mathbf{x} + \mathbf{h}\mathbf{y}, \\
& ek^{\text{PKE}} = (seed_{\text{ek}}^{\text{PKE}}, \mathbf{s}), \quad dk^{\text{PKE}} = seed_{\text{dk}}^{\text{PKE}}. \\
\text{Encrypt}(ek^{\text{PKE}}, m, \theta) \rightarrow c^{\text{PKE}} : & \mathbf{r}_1 \xleftarrow{\$} A_{L_{\text{bd}}, \gamma}^{(w_r)}, \quad \mathbf{r}_2 \xleftarrow{\$} A_{L_{\text{bd}}, \gamma}^{(w_r)}, \quad \mathbf{e} \xleftarrow{\$} \mathcal{R}_{w_e}, \\
& \mathbf{u} = \mathbf{r}_1 + \mathbf{h}\mathbf{r}_2, \quad \mathbf{v} = C.\text{Encode}(m) + \text{Truncate}(\mathbf{s}\mathbf{r}_2 + \mathbf{e}, \ell), \\
& c^{\text{PKE}} = (\mathbf{u}, \mathbf{v}). \\
\text{Decrypt}(dk^{\text{PKE}}, c^{\text{PKE}}) \rightarrow m' : & m' = C.\text{Decode}(\mathbf{v} - \text{Truncate}(\mathbf{u}\mathbf{y}, \ell)).
\end{aligned}$$

TriQ-KEM data flow

$$\begin{aligned}
\text{KeyGen}() \rightarrow (ek^{\text{KEM}}, dk^{\text{KEM}}) : & seed^{\text{KEM}} \xleftarrow{\$} \{0, 1\}^k, \quad \text{derive } (seed^{\text{PKE}}, \sigma) \text{ from } seed^{\text{KEM}}, \\
& (ek^{\text{PKE}}, dk^{\text{PKE}}) \leftarrow \text{PKE.KeyGen}(seed^{\text{PKE}}), \quad ek^{\text{KEM}} = ek^{\text{PKE}}, \\
& dk^{\text{KEM}} = (ek^{\text{KEM}}, dk^{\text{PKE}}, \sigma, seed^{\text{KEM}}). \\
\text{Encaps}(ek^{\text{KEM}}) \rightarrow (c^{\text{KEM}}, K) : & m \xleftarrow{\$} \{0, 1\}^k, \quad salt \xleftarrow{\$} \{0, 1\}^{s|salt|}, \\
& (K, \theta) = G(H(ek^{\text{KEM}}) \parallel m \parallel salt), \\
& c^{\text{PKE}} \leftarrow \text{PKE.Encrypt}(ek^{\text{KEM}}, m, \theta), \\
& c^{\text{KEM}} = (c^{\text{PKE}}, salt). \\
\text{Decaps}(dk^{\text{KEM}}, c^{\text{KEM}}) \rightarrow K_{\text{dec}} : & m' \leftarrow \text{PKE.Decrypt}(dk^{\text{PKE}}, c^{\text{PKE}}), \\
& (K', \theta') \leftarrow G(H(ek^{\text{KEM}}) \parallel m' \parallel salt), \\
& c^{\text{KEM}'} \leftarrow (\text{PKE.Encrypt}(ek^{\text{KEM}}, m', \theta'), salt), \\
& \text{output } K' \text{ if } c^{\text{KEM}'} = c^{\text{KEM}}, \\
& \text{else output } J(H(ek^{\text{KEM}}) \parallel \sigma \parallel c^{\text{KEM}}).
\end{aligned}$$

Figure 1: TriQ-PKE and TriQ-KEM data flow

III-A TriQ-PKE Algorithms

TriQ-PKE uses a decodable Reed–Solomon/Reed–Muller (RSRM) code C for message encoding and decoding. The code C has binary codeword length $n_1 n_2$ and dimension k . It is the concatenation of an outer Reed–Solomon code over $\mathbb{F}_{256}$ and an inner repeated Reed–Muller code over $\mathbb{F}_2$. The outer Reed–Solomon code has parameters

$$[n_1, k/8, n_1 - k/8 + 1]_{256}.$$

The field $\mathbb{F}_{256}$ is defined by the primitive polynomial $x^8 + x^4 + x^3 + x^2 + 1$; the RS generator polynomial is $g(x) = \prod_{i=1}^{2\delta} (x - \alpha^i)$, where δ is the RS correction capability and α is a root of the primitive polynomial. The inner code is a repeated Reed–Muller code with binary parameters

$$[n_2, 8, n_2/2]_2.$$

The concrete RSRM instances are listed in Table 3. For $m \in \mathbb{F}_2^k$, write

$$C.\text{Encode}(m) = m\mathbf{G} \in \mathbb{F}_2^{n_1 n_2},$$

where $\mathbf{G}$ is the public generator matrix of C . The decoder $C.\text{Decode}(\cdot)$ applies inner Reed–Muller decoding by the fast Hadamard transform (FHT), followed by outer Reed–Solomon decoding by the Berlekamp–Massey (BM) algorithm.

TriQ-PKE also uses a binary systematic quasi-cyclic code of length $2n$ and dimension n , where n is the smallest primitive prime greater than the RSRM codeword length $n_1 n_2$. All quasi-cyclic operations take place in the ring $\mathcal{R} = \mathbb{F}_2[X]/(X^n - 1)$. The difference between the quasi-cyclic length and the RSRM codeword length is

$$\ell = n - n_1 n_2.$$

The public element $\mathbf{h} \in \mathcal{R}$ defines the syndrome map $(\mathbf{a}, \mathbf{b}) \mapsto \mathbf{a} + \mathbf{h}\mathbf{b}$ used below to form $\mathbf{s} = \mathbf{x} + \mathbf{h}\mathbf{y}$ and $\mathbf{u} = \mathbf{r}_1 + \mathbf{h}\mathbf{r}_2$.

TriQ-PKE consists of three routines:

$$\begin{aligned} & \text{TriQ.PKE.KeyGen}(\text{seed}^{\text{PKE}}), \\ & \text{TriQ.PKE.Encrypt}(ek^{\text{PKE}}, m, \theta), \\ & \text{TriQ.PKE.Decrypt}(dk^{\text{PKE}}, c^{\text{PKE}}). \end{aligned}$$

The PKE key generation uniformly samples

$$\mathbf{x} \xleftarrow{\$} \mathcal{R}_{w_{\mathbf{x}}}, \quad \mathbf{y} \xleftarrow{\$} \mathcal{R}_{w_{\mathbf{y}}},$$

and PKE encryption samples

$$\mathbf{r}_1 \xleftarrow{\$} A_{L_{\text{bd}}, \gamma}^{(w_{\mathbf{r}})}, \quad \mathbf{r}_2 \xleftarrow{\$} A_{L_{\text{bd}}, \gamma}^{(w_{\mathbf{r}})}, \quad \mathbf{e} \xleftarrow{\$} \mathcal{R}_{w_{\mathbf{e}}}.$$

The bounded-density sampling procedure for $\mathbf{r}_1$ and $\mathbf{r}_2$ is specified in Section III-B. In all submitted instances $w_{\mathbf{x}} = w_{\mathbf{y}} = w_{\mathbf{r}} < w_{\mathbf{e}}$, so the PKE component uses unbalanced weights for the PKE encryption vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}$ together with public bounded-density sampling for $\mathbf{r}_1$ and $\mathbf{r}_2$.

Key generation. Given seed^{PKE} , TriQ.PKE.KeyGen derives $\text{seed}_{\text{dk}}^{\text{PKE}}$ and $\text{seed}_{\text{ek}}^{\text{PKE}}$ through a domain-separated derivation function. From $\text{seed}_{\text{dk}}^{\text{PKE}}$, it samples $\mathbf{x} \xleftarrow{\$} \mathcal{R}_{w_{\mathbf{x}}}$ and $\mathbf{y} \xleftarrow{\$} \mathcal{R}_{w_{\mathbf{y}}}$. From $\text{seed}_{\text{ek}}^{\text{PKE}}$, it samples $\mathbf{h} \xleftarrow{\$} \mathcal{R}$ and computes $\mathbf{s} = \mathbf{x} + \mathbf{h}\mathbf{y}$. The outputs are

$$ek^{\text{PKE}} = (\text{seed}_{\text{ek}}^{\text{PKE}}, \mathbf{s}), \quad dk^{\text{PKE}} = \text{seed}_{\text{dk}}^{\text{PKE}}.$$

Encryption. Given ek^{PKE} , $m \in \mathbb{F}_2^k$, and encryption randomness θ , the routine parses ek^{PKE} as $(\text{seed}_{\text{ek}}^{\text{PKE}}, \mathbf{s})$ and regenerates $\mathbf{h}$ from $\text{seed}_{\text{ek}}^{\text{PKE}}$. It uses θ to sample

$$\mathbf{r}_2 \xleftarrow{\$} A_{L_{\text{bd}}, \gamma}^{(w_{\mathbf{r}})}, \quad \mathbf{e} \xleftarrow{\$} \mathcal{R}_{w_{\mathbf{e}}}, \quad \mathbf{r}_1 \xleftarrow{\$} A_{L_{\text{bd}}, \gamma}^{(w_{\mathbf{r}})}.$$

It then computes

$$\mathbf{u} = \mathbf{r}_1 + \mathbf{h}\mathbf{r}_2, \quad \mathbf{v} = C.\text{Encode}(m) + \text{Truncate}(\mathbf{s}\mathbf{r}_2 + \mathbf{e}, \ell).$$

The output is

$$c^{\text{PKE}} = (\mathbf{u}, \mathbf{v}).$$

Decryption. Given dk^{PKE} and $c^{\text{PKE}} = (\mathbf{u}, \mathbf{v})$, the routine regenerates $\mathbf{y}$ from $dk^{\text{PKE}} = \text{seed}_{\text{dk}}^{\text{PKE}}$, computes

$$\mathbf{v} - \text{Truncate}(\mathbf{u}\mathbf{y}, \ell) = C.\text{Encode}(m) + \text{Truncate}(\mathbf{x}\mathbf{r}_2 + \mathbf{r}_1\mathbf{y} + \mathbf{e}, \ell),$$

and applies the RSRM decoder. The output is a candidate message $m' \in \mathbb{F}_2^k$.

The decryption noise before truncation is

$$\mathbf{e}' = \mathbf{x}\mathbf{r}_2 + \mathbf{r}_1\mathbf{y} + \mathbf{e}.$$

Section V analyzes and bounds decoding failure induced by this effective error vector for honest encryption and for the density-interval attack model.

III-B Bounded-Density Rejection Rule and Vector Sampling

TriQ-PKE samples $\mathbf{r}_1$ and $\mathbf{r}_2$ by drawing fixed-weight candidates and applying a public local-density condition. The condition is public: for each candidate vector, it depends only on the candidate support and the public parameters L_{bd} and γ , and not on secret data. For $a \in \mathbb{Z}_n$ and $1 \leq L \leq n$, let

$$I_L(a) = \{a, a+1, \dots, a+L-1\} \pmod{n}$$

be a cyclic interval of length L . For $0 \leq \gamma \leq L$, define

$$A_{L,\gamma}^{(w)} = \left\{ \mathbf{v} \in \mathcal{R}_w : \max_{a \in \mathbb{Z}_n} \text{wt}(\mathbf{v}|_{I_L(a)}) \leq \gamma \right\}. \quad (1)$$

The sampler first draws a fixed-weight candidate uniformly from $\mathcal{R}_w$, tests the bounded-density condition, and repeats until a candidate is accepted or until an upper bound on attempts is reached. Conditioned on acceptance before the attempt limit, its output is uniform over $A_{L_{\text{bd}},\gamma}^{(w)}$.

The bounded-density check itself is constant-time: it computes the maximum window count across all support positions using branch-free arithmetic, so its cost depends only on the weight w and not on the support position or density. This property is preserved in both the reference and the optimized implementations.

The following lemma bounds the rejection probability of the sampler.

Let $\mathbf{v}$ be uniform over $\mathcal{R}_w$. Define

$$p_{\text{rej}}^{(w)} = \Pr[\mathbf{v} \notin A_{L,\gamma}^{(w)}], \quad q_{\text{bd}}^{(w)} = 1 - p_{\text{rej}}^{(w)}, \quad \lambda_{\text{bd}}^{(w)} = -\log_2 q_{\text{bd}}^{(w)}.$$

Lemma III.1 (Bound for bounded-density rejection). *For $1 \leq L \leq n$, $1 \leq \gamma < w$, and $\gamma n > w(L-1)$, the rejection probability satisfies*

$$p_{\text{rej}}^{(w)} \leq U_{\text{clump}}(n, w, L, \gamma), \quad (2)$$

where

$$U_{\text{clump}}(n, w, L, \gamma) = \min \left\{ 1, \frac{w}{\binom{n-1}{w-1}} \sum_{a=\gamma-1}^{L-2} \binom{a-1}{\gamma-2} \left(\binom{n-L}{w-\gamma} - \binom{n-2L+1+a}{w-\gamma} \right) \right\}.$$

The concrete values of L_{bd} , γ , N_{max} , and the corresponding rejection bounds are listed in Section IV.

III-C TriQ-KEM Algorithms

TriQ-KEM applies the Fujisaki–Okamoto transform to TriQ-PKE. The transform uses domain-separated functions H , G , and J . The function H hashes the encapsulation key, G derives the shared secret K and the PKE encryption randomness θ from $H(ek^{\text{KEM}})$, the message representative, and the salt, and J derives the implicit-rejection key $\bar{K}$ from $H(ek^{\text{KEM}})$, σ , and the KEM ciphertext. The three derivations use separate domains. The KEM seed is expanded to obtain seed^{PKE} and σ , while the PKE seed derivation obtains $\text{seed}_{\text{dk}}^{\text{PKE}}$ and $\text{seed}_{\text{ek}}^{\text{PKE}}$ from seed^{PKE} .

The three KEM routines are

$$\begin{aligned} & \text{TriQ.KEM.KeyGen}(), \\ & \text{TriQ.KEM.Encaps}(ek^{\text{KEM}}), \\ & \text{TriQ.KEM.Decaps}(dk^{\text{KEM}}, c^{\text{KEM}}). \end{aligned}$$

KEM key generation. TriQ.KEM.KeyGen samples

$$seed^{\text{KEM}} \xleftarrow{\$} \{0, 1\}^k.$$

The seed $seed^{\text{KEM}}$ is used to derive $seed^{\text{PKE}}$ and σ through the domain-separated derivation functions. It then runs

$$(ek^{\text{PKE}}, dk^{\text{PKE}}) \leftarrow \text{TriQ.PKE.KeyGen}(seed^{\text{PKE}}),$$

sets $ek^{\text{KEM}} = ek^{\text{PKE}}$, and forms

$$dk^{\text{KEM}} = (ek^{\text{KEM}}, dk^{\text{PKE}}, \sigma, seed^{\text{KEM}}).$$

Here $ek^{\text{KEM}} = ek^{\text{PKE}} = (seed_{\text{ek}}^{\text{PKE}}, \mathbf{s})$ and $dk^{\text{PKE}} = seed_{\text{dk}}^{\text{PKE}}$.

Encapsulation. Given ek^{KEM} , TriQ.KEM.Encaps samples

$$m \xleftarrow{\$} \{0, 1\}^k, \quad salt \xleftarrow{\$} \{0, 1\}^{8|salt|}.$$

It computes

$$(K, \theta) \leftarrow G(H(ek^{\text{KEM}}) \parallel m \parallel salt),$$

runs

$$c^{\text{PKE}} \leftarrow \text{TriQ.PKE.Encrypt}(ek^{\text{KEM}}, m, \theta),$$

and outputs

$$c^{\text{KEM}} = (c^{\text{PKE}}, salt), \quad K \in \{0, 1\}^k.$$

Decapsulation. Given dk^{KEM} and c^{KEM} , the routine parses

$$dk^{\text{KEM}} = (ek^{\text{KEM}}, dk^{\text{PKE}}, \sigma, seed^{\text{KEM}})$$

and $c^{\text{KEM}} = (c^{\text{PKE}}, salt)$. It computes

$$m' \leftarrow \text{TriQ.PKE.Decrypt}(dk^{\text{PKE}}, c^{\text{PKE}}),$$

then recomputes

$$(K', \theta') \leftarrow G(H(ek^{\text{KEM}}) \parallel m' \parallel salt)$$

and

$$c^{\text{PKE}'} \leftarrow \text{TriQ.PKE.Encrypt}(ek^{\text{KEM}}, m', \theta').$$

Let $c^{\text{KEM}'} = (c^{\text{PKE}'}, salt)$, and compute

$$\bar{K} \leftarrow J(H(ek^{\text{KEM}}) \parallel \sigma \parallel c^{\text{KEM}}).$$

If $c^{\text{KEM}'} = c^{\text{KEM}}$, the output is K' ; otherwise the output is $\bar{K}$.

IV Parameter Sets and Sizes

This section lists the submitted TriQ-KEM instances and sizes. Detailed DFR, CDFR, and ISD analyses are given in Section V; runtime comparisons are given in Section VI.

IV-A Parameter Selection

The TriQ-KEM parameter sets are selected under two design constraints: security strength and decryption-failure behavior.

Security strength. The submitted instances target the classical security levels

$$\lambda \in \{128, 256, 384, 512\},$$

with quantum security targets 80, 128, 192, 256 bits.

The security of the underlying quasi-cyclic syndrome decoding problems is evaluated through the best known classical attacks, information-set decoding (ISD), and through quantum-Prange bit-complexity estimates. Section V-C gives the detailed analysis.

For a given parameter set, let

$$\underline{T}_{\text{ISD}} = \min \{ \underline{T}_{\text{ISD}}(2n, n, (\mathbf{x}, \mathbf{y})), \underline{T}_{\text{ISD}}(3n, n, (\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})) \}$$

be the smaller of the two structured-ISD lower bounds, and let $\underline{T}_{\text{Q}}$ be the corresponding minimal quantum-Prange bit-complexity estimate. The requirements are $\underline{T}_{\text{ISD}} > \lambda$; the quantum-Prange bit-complexity estimate $\underline{T}_{\text{Q}}$ is reported in the security summary. The bounded-density correction to the bounded-density-conditioned 3-CW-QCSD estimate is included in the estimate (Section V-C).

Decryption-failure constraints. Because the density-interval attack of Guo and Johansson (Section V) can amplify failures through repeated weak-ciphertext queries, TriQ-KEM controls the attack-aware cumulative decryption-failure probability (CDFR) under a bounded-query budget $q_d \leq 2^{80}$. As a comparison calculation, applying the same bounded-query CDFR evaluation to the Kyber parameter sets under their failure-boosting heuristic gives cumulative failure exponents of about 11–27 bits for classical search and 16–18 bits for quantum search. The target upper bounds for the post-rejection CDFR are

$$2^{-32}, 2^{-64}, 2^{-96}, 2^{-128}$$

for the four instances, respectively. The post-rejection CDFR is the attack-aware decryption-failure metric used for parameter selection; the natural DFR is the honest-encapsulation correctness reference. The achieved bounds are summarized in Section V.

IV-B Parameter Sets

TriQ-KEM uses an RSRM code of length $n_1 n_2$ and dimension k , where n_1 is the outer RS length and n_2 the inner repeated RM length. The ring length n is the smallest primitive prime above $n_1 n_2$. The secret vectors $(\mathbf{x}, \mathbf{y})$ have weights $w_{\mathbf{x}}, w_{\mathbf{y}}$, the encryption vectors use weight $w_{\mathbf{r}}$ for $(\mathbf{r}_1, \mathbf{r}_2)$ and $w_{\mathbf{e}}$ for $\mathbf{e}$, with $w_{\mathbf{x}} = w_{\mathbf{y}} = w_{\mathbf{r}}$.

Table 2: TriQ-KEM parameter sets

Instance	n_1	n_2	$n_1 n_2$	n	k	$w_{\mathbf{x}} = w_{\mathbf{y}} = w_{\mathbf{r}}$	$w_{\mathbf{e}}$
TriQ-KEM-128	42	384	16128	16301	128	67	106
TriQ-KEM-256	64	768	49152	50363	256	132	234
TriQ-KEM-384	108	896	96768	97651	384	196	350
TriQ-KEM-512	136	1152	156672	157627	512	260	475

The corresponding RSRM code instances are listed in Table 3.

Table 3: RSRM code instances used by TriQ-KEM

Instance	Outer RS code	RS correction capability	Inner RM code
TriQ-KEM-128	$[42, 16, 27]_{256}$	13	$[384, 8, 192]_2$
TriQ-KEM-256	$[64, 32, 33]_{256}$	16	$[768, 8, 384]_2$
TriQ-KEM-384	$[108, 48, 61]_{256}$	30	$[896, 8, 448]_2$
TriQ-KEM-512	$[136, 64, 73]_{256}$	36	$[1152, 8, 576]_2$

TriQ-KEM additionally applies bounded-density rejection sampling to $\mathbf{r}_1$ and $\mathbf{r}_2$. For each parameter set, the rejection rule uses a single cyclic window length L_{bd} and accepts only vectors in $A_{L_{\text{bd}}, \gamma}^{(w_r)}$. The selected rejection parameters and the corresponding rejection-probability upper bounds from Lemma III.1 are listed in Table 4.

Table 4: Single-scale bounded-density rejection parameters for $\mathbf{r}_1$ and $\mathbf{r}_2$

Instance	L_{bd}	γ	N_{max}	U_{clump}	$\lambda_{\text{bd}}^{(w_r)}$
TriQ-KEM-128	384	7	256	0.04161	0.0613
TriQ-KEM-256	768	9	256	0.01727	0.0251
TriQ-KEM-384	896	9	256	0.01248	0.0181
TriQ-KEM-512	1152	9	256	0.02551	0.0373

For all four instances, the expected number of attempts for one bounded-density sampling call is at most $1/(1 - U_{\text{clump}}) \leq 1.044$, using the largest U_{clump} value in Table 4. Under the same bound and independent candidate draws, the probability that all 256 trials fail in one sampling call is at most U_{clump}^{256} ; for the two bounded-density samples $\mathbf{r}_1$ and $\mathbf{r}_2$ in one encryption, the corresponding union bound is $2U_{\text{clump}}^{256}$.

IV-C Key and Ciphertext Sizes

The message representative m , the implicit-rejection secret σ , and the final shared secret K have size $k/8$ bytes. TriQ-KEM represents regenerable key material by seeds. The KEM byte sizes are determined by the following formulas:

$$|ek^{\text{KEM}}| = |\text{seed}| + \left\lceil \frac{n}{8} \right\rceil, \quad |dk^{\text{PKE}}| = |\text{seed}|,$$

$$|dk^{\text{KEM}}| = |ek^{\text{KEM}}| + |dk^{\text{PKE}}| + \frac{k}{8} + |\text{seed}|,$$

$$|c^{\text{KEM}}| = \left\lceil \frac{n}{8} \right\rceil + \left\lceil \frac{n_1 n_2}{8} \right\rceil + |\text{salt}|, \quad |K| = \frac{k}{8} \text{ B},$$

where each stored seed value has length $|\text{seed}| = k/8$ B and $|\text{salt}| = 32$ B. The final $|\text{seed}|$ term in $|dk^{\text{KEM}}|$ accounts for an additional compact key seed stored by the decapsulation key. Table 5 lists the corresponding sizes.

Table 5: TriQ-KEM key, ciphertext, and shared-secret byte sizes

Instance	$ ek^{\text{KEM}} $	$ dk^{\text{KEM}} $	$ c^{\text{KEM}} $	$ K $
TriQ-KEM-128	2054	2102	4086	16
TriQ-KEM-256	6328	6424	12472	32
TriQ-KEM-384	12255	12399	24335	48
TriQ-KEM-512	19768	19960	39320	64

V Security Analysis

This section develops the attack-aware decryption-failure analysis in four steps: (i) the density-interval attack, (ii) the adversary’s query budget, (iii) the induced noise model and the resulting DFR upper bound, and (iv) the concrete CDFR evaluation for the submitted parameter sets.

V-A Security Statement

For the listed parameter instances, TriQ-KEM satisfies the following security properties:

- **Correctness:** for ciphertexts produced by the honest encapsulation algorithm, TriQ-KEM decapsulation recovers the encapsulated shared secret whenever the underlying TriQ-PKE decryption recovers the message representative. The single-query KEM correctness-failure probability is bounded by the natural DFR of the underlying PKE decryption.
- **Attack-aware CDFR:** the decryption-failure security criterion is the cumulative decryption-failure probability (CDFR) against the density-interval attack, which accounts for failure amplification over multiple decapsulation queries. The bounded-density rejection rule gives the post-rejection CDFR values reported in Table 8.
- **Underlying problems:** the concrete hardness estimates use the 2-CW-QCSD problem for public-key recovery and the 3-CW-QCSD problem, conditioned on bounded-density rejection, for ciphertext-randomness recovery. The entropy loss from bounded-density rejection is included in the ISD estimates. The corresponding classical ISD and quantum-Prange bit-complexity estimates are given in Section V-C.
- **IND-CCA2 security:** TriQ-KEM applies the Fujisaki–Okamoto transform with implicit rejection [3,4]. The IND-CCA2 security follows from the FO reduction, given the IND-CPA security of TriQ-PKE (supported by the 2-CW-QCSD and bounded-density conditioned 3-CW-QCSD estimates) and the decryption-failure analysis below. The natural DFR gives the honest-encapsulation correctness bound, while the post-rejection CDFR gives the attack-aware density-interval failure bound used for the submitted parameter sets.

Table 6 summarizes the resulting security levels and failure bounds for the four submitted instances. The classical ISD column reports the minimum over the ISD variants and over the two CW-QCSD instances; the quantum-Prange column gives the corresponding minimum bit-complexity estimate. The post-rejection CDFR is the decryption-failure metric used for the submitted parameter sets, and the natural DFR is the single-query correctness-failure bound for honest encapsulations.

Table 6: Security levels and failure bounds for TriQ-KEM

Instance	classical ISD	min Q-Prange	post-rejection CDFR	natural DFR
TriQ-KEM-128	146.67	95.18	$\leq 2^{-34.32}$	$\leq 2^{-121.90}$
TriQ-KEM-256	277.04	163.49	$\leq 2^{-85.63}$	$\leq 2^{-173.35}$
TriQ-KEM-384	404.13	229.43	$\leq 2^{-124.93}$	$\leq 2^{-209.57}$
TriQ-KEM-512	532.06	294.84	$\leq 2^{-146.56}$	$\leq 2^{-230.00}$

V-B Decryption Failure Rate Analysis

This subsection analyzes the decryption failure rate under the density-interval attack: it defines the attack domain, bounds the number of admissible decryption queries, derives the PEB-BSC DFR upper bound, and evaluates the resulting CDFR for the specified parameter sets.

V-B.1 Density-Interval Attack

The density-interval attack of Guo and Johansson [2] is an adaptive decryption-failure attack that induces elevated failure probabilities for selected ciphertexts. The adversary exploits decryption success/failure outcomes to extract failure-conditioned coordinate biases, thereby revealing information about the support of the secret vector. The attack relies on the structure of $\mathbf{e}'$ and

on BM-FHT decoding behavior for RSRM codes. In this setting, failures depend not only on total error weight but also on how the errors are distributed across RM components. Even when $\mathbf{e}'$ has only moderate total weight, a highly concentrated error pattern can still increase the probability of BM-FHT decoding failure.

The adversary searches for $\mathbf{r}_2$ containing a *dense interval* of length l_1 with l_0 ones and density $l_0/l_1 > 1/2$. Write

$$\mathbf{r}_2 = \mathbf{r}_2^1 + \mathbf{r}_2^2, \quad \mathbf{r}_2^2 = (0, \dots, 0, \mathbf{b}, 0, \dots, 0),$$

where $\mathbf{b} \in \mathbb{F}_2^{l_1}$ has $\text{wt}(\mathbf{b}) = l_0$. Then

$$\mathbf{e}' = (\mathbf{x}\mathbf{r}_2^1 + \mathbf{y}\mathbf{r}_1 + \mathbf{e}) + \mathbf{x}\mathbf{r}_2^2,$$

and the term $\mathbf{x}\mathbf{r}_2^2$ creates concentrated error contributions over RM blocks. Since $w_{\mathbf{x}} \ll n$, each nonzero position of $\mathbf{x}$ shifts the dense interval. Thus $\mathbf{x}\mathbf{r}_2^2$ can be viewed as a sum of about $w_{\mathbf{x}}$ translated dense intervals. This makes the blockwise error distribution more concentrated and increases the probability of BM-FHT decoding failure.

Let $\mathcal{D}$ denote the event that decryption fails, and let e_i be the i -th component of $\mathbf{e}$ (known to the adversary for each query). The key observation is a failure-conditioned bias: indices i inside the dense interval have a different conditional frequency $\Pr[e_i = 1 \mid \mathcal{D}]$ from indices outside. This yields a distinguisher based on $\Pr[e_i = 1 \mid \mathcal{D}]$. The distinguisher localizes the dense interval in $\mathbf{e}'$ and then leaks information about the support of the secret $\mathbf{x}$.

This attack consists of three stages [2]:

1. **Precomputation.** The adversary precomputes a set $\mathcal{S}$ of triples $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$ and related ciphertexts such that $\mathbf{r}_2$ contains a length- l_1 interval with at least l_0 ones. The size of $\mathcal{S}$ does not exceed the decryption query limit $q = 2^{80}$.
2. **Decryption queries.** For each $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}) \in \mathcal{S}$, the adversary queries the decryption oracle, recording whether $\mathcal{D}$ occurs.
3. **Private-key reconstruction.** From the subset of queries that fail, the adversary estimates failure-conditioned coordinate frequencies of the known error vector and applies a sliding-window statistic over candidate intervals. Repeated biased interval positions can then be correlated with shifts induced by nonzero positions of $\mathbf{x}$, leaking information about the support of $\mathbf{x}$. Guo and Johansson reported that roughly 2^{16} failure ciphertexts are required for private-key recovery in their analysis [2].

The attack regime considered below requires high interval density, typically $l_0/l_1 > 1/2$. In the enumeration, we restrict to integer pairs satisfying $1 \leq l_0 \leq \min\{w_{\mathbf{r}}, l_1\}$ and $l_0/l_1 > 1/2$, equivalently $l_0 \leq l_1 < 2l_0$. The next step is to translate this density condition into an attack-cost bound.

V-B.2 Attack Complexity Bound

The density-interval attack [2] is relevant only when the adversary can generate ciphertexts that trigger decryption failures. We therefore bound the attack cost by the ISD security estimate $\underline{T}_{\text{ISD}}$ used for the relevant CW-QCSD problems.

The event that a uniformly random $\mathbf{r}_2$ contains a length- l_1 cyclic window with at least l_0 nonzero coordinates is precisely the event that $\mathbf{r}_2$ is rejected by the bounded-density condition of Section III-B with window length $L = l_1$ and bound $\gamma = l_0 - 1$. Indeed, $\mathbf{r}_2 \notin A_{l_1, l_0-1}^{(w_{\mathbf{r}})}$ means that some cyclic interval of length l_1 contains at least $(l_0 - 1) + 1 = l_0$ support points, which is exactly the dense-interval condition required by the attack. Lemma III.1 therefore gives a direct upper bound on the probability of finding such a vector. With U_{clump} defined in Eq. (2), the

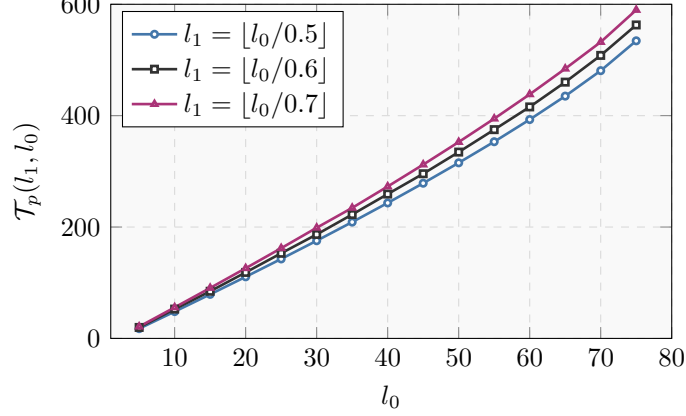


Figure 2: Precomputation complexity $\mathcal{T}_p(l_1, l_0)$ as a function of attack parameters l_1 and l_0 (TriQ-KEM-128 parameter set)

expected number of trials per weak ciphertext is at least $1/U_{\text{clump}}(n, w_{\mathbf{r}}, L = l_1, \gamma = l_0 - 1)$, and the corresponding bit-complexity lower bound is

$$\mathcal{T}_p(l_1, l_0) = \max \left\{ 0, -\log_2 U_{\text{clump}}(n, w_{\mathbf{r}}, L = l_1, \gamma = l_0 - 1) \right\}, \quad (3)$$

i.e., at least $2^{\mathcal{T}_p(l_1, l_0)}$ trials per weak ciphertext on average.

Fig. 2 shows the precomputation cost $\mathcal{T}_p(l_1, l_0)$ as a function of l_1 and l_0 . For fixed n and $w_{\mathbf{r}}$, this cost increases with l_0 and with the density l_0/l_1 . Imposing the budget constraint $\mathcal{T}_p(l_1, l_0) \leq \underline{T}_{\text{ISD}}$ therefore determines the feasible search space for (l_1, l_0) . For the four TriQ-KEM parameter sets and under the restriction $l_0/l_1 > 1/2$, the feasible values of l_1 remain much smaller than n .

According to Eq. (3), the cost of generating a weak ciphertext with parameters (l_1, l_0) is lower-bounded by $2^{\mathcal{T}_p(l_1, l_0)}$. Given that the adversary's total computational budget is limited by $\underline{T}_{\text{ISD}}$, the number of decryption queries it can make is upper-bounded by

$$q(l_1, l_0) = \min \left\{ 2^{80}, \left\lfloor 2^{\underline{T}_{\text{ISD}} - \mathcal{T}_p(l_1, l_0)} \right\rfloor \right\}.$$

The cap 2^{80} follows the decapsulation-query bound used in the Kyber failure analysis [5]; TriQ-KEM applies the same query bound to the Guo density-interval attack together with the CW-QCSD-based ISD cost bound above.

Let $\text{DFR}(l_1, l_0)$ denote the DFR for weak ciphertexts generated with parameters (l_1, l_0) . Then, the probability of observing at least one failure is $1 - (1 - \text{DFR}(l_1, l_0))^{q(l_1, l_0)}$. We define the **cumulative decryption-failure probability (CDFR)** as

$$\text{CDFR} = \max_{(l_1, l_0)} \left\{ 1 - (1 - \text{DFR}(l_1, l_0))^{q(l_1, l_0)} \right\}, \quad (4)$$

where the maximum is taken over feasible density-interval parameters. The CDFR criterion is $\text{CDFR} < \varepsilon$ for the stated target ε . Under this criterion, even an adversary with resources comparable to the relevant CW-QCSD ISD estimates triggers a decryption failure with probability below ε .

Remark V.1. The CDFR estimate below is restricted to the density-interval attack model analyzed here, with $l_0/l_1 > 1/2$ and $l_1 < n_2$. If an adversary instead uses intervals with density $l_0/l_1 < 1/2$, then the distinguisher based on failure-conditioned coordinate statistics no longer directly applies.

V-B.3 Noise Model

To upper-bound the DFR for weak ciphertexts, we first approximate the non-dense part of the decryption noise by its coordinate crossover probability.

Lemma V.2 (Coordinate probability for a product of fixed-weight vectors). *Let $\mathbf{x}$ and $\mathbf{r}$ be independent uniform elements of $\mathcal{R}_{w_x}$ and $\mathcal{R}_{w_r}$, respectively, and set $\mathbf{z} = \mathbf{x}\mathbf{r}$. For any coordinate k ,*

$$\Pr[z_k = 1] = \sum_{\substack{1 \leq \ell \leq \min(w_x, w_r) \\ \ell \text{ odd}}} \frac{\binom{w_x}{\ell} \binom{n-w_x}{w_r-\ell}}{\binom{n}{w_r}}. \quad (5)$$

Lemma V.3 (BSC crossover probability for the TriQ-PKE decryption noise). *For a fixed coordinate k , let*

$$U = (\mathbf{x}\mathbf{r}_2)_k, \quad V = (\mathbf{r}_1\mathbf{y})_k, \quad W = e_k,$$

and let $p_U = \Pr[U = 1]$, $p_V = \Pr[V = 1]$, and $p_W = \Pr[W = 1]$, with p_U and p_V computed by Eq. (5). For

$$\mathbf{e}' = \mathbf{x}\mathbf{r}_2 + \mathbf{r}_1\mathbf{y} + \mathbf{e},$$

the coordinate probability is

$$\Pr[e'_k = 1] = p_U + p_V + p_W - 2p_U p_V - 2p_U p_W - 2p_V p_W + 4p_U p_V p_W.$$

Under the density-interval attack, the induced noise can be written as

$$\mathbf{e}' = \mathbf{x}\mathbf{r}_2^1 + \mathbf{y}\mathbf{r}_1 + \mathbf{e} + \mathbf{x}\mathbf{r}_2^2,$$

where $\mathbf{r}_2^2$ corresponds to a dense interval with parameters (l_1, l_0) .

It remains to model how this noise enters RSRM decoding. An RSRM codeword $\mathbf{c}$ is partitioned into n_1 segments, each being an RM codeword of length n_2 , i.e., $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_{n_1})$ with $\mathbf{c}_i \in \mathbb{F}_2^{n_2}$. The noise vector $\mathbf{e}' = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{n_1}, \mathbf{e}_{\text{trun}})$ consists of the error terms, where $\mathbf{e}_i \in \mathbb{F}_2^{n_2}$ for $i = 1, \dots, n_1$ and $\mathbf{e}_{\text{trun}} \in \mathbb{F}_2^{n-n_1n_2}$ corresponds to the truncated part of the ciphertext. Thus, the codeword to be decoded is given by $(\mathbf{c}_1 + \mathbf{e}_1, \dots, \mathbf{c}_{n_1} + \mathbf{e}_{n_1})$. Decoding proceeds in two steps:

- Each segment $\mathbf{c}_i + \mathbf{e}_i$ is decoded by the FHT decoder to obtain a byte message $\mathbf{m}_i \in \mathbb{F}_2^8$;
- The results are assembled into an RS codeword $(\mathbf{m}_1, \dots, \mathbf{m}_{n_1})$, which is then decoded using the BM decoder.

The DFR of the RSRM code depends not only on the total weight of $\mathbf{e}'$, but also on the distribution of errors across the individual $\mathbf{e}_i$.

Because a dense interval may span two adjacent $\mathbf{e}_i$, direct DFR computation would involve complicated conditional probabilities. To keep the DFR analysis tractable, the following conservative approximation method is used.

1. Since all feasible $l_1 \ll n$ for TriQ-KEM parameters, we model $\mathbf{x}\mathbf{r}_2^1 + \mathbf{y}\mathbf{r}_1 + \mathbf{e}$ as a BSC with error probability p as described in Lemma V.3. This follows the coordinate-wise BSC modeling approach used in refined HQC DFR analyses [1]; in the present calculation it is used as a conservative approximation for the non-dense part of the TriQ-KEM noise.
2. We assume that $\mathbf{x}\mathbf{r}_2^2$ consists of w_x dense intervals, which are randomly distributed across the n_1 segments $(\mathbf{e}_1, \dots, \mathbf{e}_{n_1})$, with each dense interval confined to a single $\mathbf{e}_i$. This assumption treats the DFRs of different RM codes as uncorrelated and assumes no dense intervals occur in $\mathbf{e}_{\text{trun}}$.

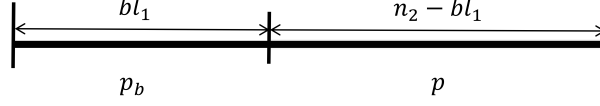


Figure 3: Error rate representation for $\mathbf{e}_i$ containing b dense intervals

3. A fixed parameter set $\mathcal{M} = \{(\zeta_b, b), 0 \leq b \leq n_1\}$ is used to fit the distribution of dense intervals across the $\mathbf{e}_i$, where ζ_b denotes the number of $\mathbf{e}_i$ containing b dense intervals, satisfying $\sum \zeta_b = n_1$ and $\sum \zeta_b \cdot b = w_{\mathbf{x}}$. Following the PEB-BSC methodology of [6], this occupancy profile is obtained from a Poisson approximation with mean $\mu = w_{\mathbf{x}}/n_1$, followed by integer adjustment so that the two constraints above hold exactly.
4. Each dense interval in $\mathbf{x}\mathbf{r}_2^2$ is treated as a region with error probability $p_l = l_0/l_1$, where the effective error probability inside such an interval in $\mathbf{e}'$ becomes

$$p_b = p_l(1 - p) + p(1 - p_l).$$

For an $\mathbf{e}_i$ containing b dense intervals:

- If $bl_1 \leq n_2$, we assume the b intervals do not overlap.
- If $bl_1 > n_2$, we approximate the error rate across $\mathbf{e}_i$ as p_b .

Since $p_l > 0.5$, this approach generally overestimates the weight of $\mathbf{e}_i$. The distribution of error rates across the segments of $\mathbf{e}_i$ is illustrated in Fig. 3.

This simplified noise model is referred to as the **Poisson-Fitted Error Block-BSC**, abbreviated as **PEB-BSC**. We use the PEB-BSC model described above as a conservative approximation for the DFR calculation under the density-interval attack.

V-B.4 DFR Upper Bound

The final DFR upper bound is then computed under this conservative approximation.

For a binary code C , let $P_{\text{fail}}(C, w)$ be the conditional probability that the maximum-likelihood decoder fails on a uniformly random error of weight w . The corresponding weight-wise failure profile is

$$\mathbf{P}_{\text{fail}}(C) = (P_{\text{fail}}(C, 1), \dots, P_{\text{fail}}(C, n)). \quad (6)$$

For a channel with error-weight distribution $\Pr(\text{wt}(\mathbf{e}) = w)$, the decoding-failure rate is bounded by

$$P_e \leq \sum_{w=1}^n P_{\text{fail}}(C, w) \cdot \Pr(\text{wt}(\mathbf{e}) = w). \quad (7)$$

Theorem V.4 (Upper Bound on Weight-Wise Failure Probability for RM Codes, [1]). *The weight-wise failure probability $P_{\text{fail}}(C, w)$ of an RM code with dimension 8 and minimum distance d satisfies*

$$P_{\text{fail}}(C, w) \leq \min \left\{ 1, E_w / \binom{n}{w} \right\},$$

$$\text{where } E_w = \frac{255}{2} \binom{d}{d/2} \binom{d}{w-d/2} + 255 \sum_{j=d/2+1}^d \binom{d}{j} \binom{d}{w-j} + \frac{1}{2} (255)^2 \sum_{j=0}^{d/2} \binom{d/2}{j} \binom{d/2}{w-d+j}.$$

Theorem V.4 is used as the formal weight-wise RM failure upper bound throughout the DFR and CDFR analysis. Hence, for a given RM code, determining the blockwise weight distribution of the error suffices to upper-bound the RM decoding-failure probability.

RSRM DFR Bound in PEB-BSC. Under the PEB-BSC model, for $\mathbf{e}_i$ containing b dense intervals, the probability that $\text{wt}(\mathbf{e}_i) = w$ is given as follows. If $b \cdot l_1 \leq n_2$, then $\Pr(\text{wt}(\mathbf{e}_i) = w)$ is

$$\sum_{w_1=\max(0, w-(n_2-b \cdot l_1))}^{\min(b \cdot l_1, w)} \binom{b \cdot l_1}{w_1} p_b^{w_1} (1-p_b)^{b \cdot l_1 - w_1} \cdot \binom{n_2 - b \cdot l_1}{w - w_1} p^{w-w_1} (1-p)^{n_2 - b \cdot l_1 - w + w_1}. \quad (8)$$

If $b \cdot l_1 \geq n_2$, we have

$$\Pr(\text{wt}(\mathbf{e}_i) = w) = \binom{n_2}{w} p_b^w (1-p_b)^{n_2-w}. \quad (9)$$

Combined with Theorem V.4, the DFR of the RM code is bounded as follows.

Theorem V.5 (Upper Bound on the DFR of RM Code in PEB-BSC). *Let C be an RM code with parameters $[128s, 8, 64s]_2$. Let $P_{\text{fail}}(C, w)$ be the HQC weight-wise failure upper bound from Theorem V.4. If blockwise error $\mathbf{e}_i$ contains b dense intervals, then the DFR of the RM code C satisfies*

$$p_{\text{rm}}(b) \leq \sum_{w=1}^{n_2} P_{\text{fail}}(C, w) \cdot \Pr(\text{wt}(\mathbf{e}_i) = w). \quad (10)$$

Since the DFR of the RM codes depends only on the error weight of $\mathbf{e}_i$, in the PEB-BSC, we approximate that the different inner RM codes of the RSRM code are independent events. The number of decoding failures X_b among the ζ_b RM codes containing b dense intervals satisfies $X_b \sim \text{Binomial}(\zeta_b, p_{\text{rm}}(b))$. Thus, the total number of decoding-failed RM codes $e_{\text{rm}} = \sum_{b=0}^m X_b$ can be seen as the sum of $m = \#\mathcal{M}$ Binomial variables X_b . The DFR of the RSRM code equals the probability that e_{rm} exceeds the error correction capability of the RS code.

Theorem V.6 (Upper Bound on the DFR of the RSRM Code in PEB-BSC). *For an RSRM code with RS code parameters $[n_1, k/8, n_1 - k/8 + 1]_{256}$, where k is the binary plaintext length, its DFR in PEB-BSC does not exceed*

$$\Pr(e_{\text{rm}} > t) = 1 - \sum_{j=0}^t \text{coeff} \left(\prod_{b=0}^m (1 - p_{\text{rm}}(b) + p_{\text{rm}}(b)z)^{\zeta_b}, z^j \right),$$

where $t = \lfloor \frac{n_1 - k/8}{2} \rfloor$, and $\text{coeff}(P(z), z^j)$ denotes the coefficient of z^j in polynomial $P(z)$.

Proof location. See Appendix A-B.2.

For any attack parameters (l_1, l_0) , the above procedure yields the DFR upper bound used in the CDFR evaluation under the PEB-BSC conservative approximation.

V-B.5 CDFR Evaluation

The specified TriQ-KEM parameter sets in Section IV give the following attack-aware cumulative decryption-failure estimates. The pre-rejection tables are reference calculations before applying the public bounded-density rejection rule; the post-rejection tables give the CDFR values used for the submitted parameter sets.

For each parameter set, we exhaustively enumerate all feasible density-interval parameters (l_1, l_0) satisfying the feasibility conditions of the attack model, and evaluate the corresponding attack-aware DFR and cumulative trigger probability under the specified bounded-query setting $q_d \leq 2^{80}$. The maximizing pair in the classical setting is reported in Table 7. This table gives the same attack-domain evaluation before the bounded-density rejection rule is imposed and is included for comparison.

Table 7: Pre-rejection reference CDFR for the specified TriQ-KEM parameter sets under the density-interval attack [2] in the classical setting

Parameter set	T_{ISD}	(l_1, l_0)	$\mathcal{T}_p(l_1, l_0)$	$\text{DFR}(l_1, l_0)$	CDFR
TriQ-KEM-128	146.67	(25, 13)	69.54	$\leq 2^{-99.54}$	$\leq 2^{-22.42}$
TriQ-KEM-256	277.04	(61, 31)	199.12	$\leq 2^{-109.78}$	$\leq 2^{-31.86}$
TriQ-KEM-384	404.13	(93, 47)	324.93	$\leq 2^{-120.79}$	$\leq 2^{-41.59}$
TriQ-KEM-512	532.06	(123, 62)	449.17	$\leq 2^{-120.65}$	$\leq 2^{-40.65}$

With the bounded-density rejection rule of Table 4, the original density-interval search is further restricted. In the Guo density-interval regime considered here, the feasible intervals satisfy $l_1 < L_{\text{bd}}$ for all four TriQ-KEM parameter sets. Therefore, any candidate with $l_0 > \gamma$ is rejected, since a length- L_{bd} cyclic window can cover the dense interval. The post-rejection search is evaluated over

$$l_0/l_1 > 1/2, \quad l_0 \leq l_1 < 2l_0, \quad l_1 < L_{\text{bd}}, \quad l_0 \leq \gamma.$$

For accepted candidates, conditioning on the accepted sampling space can only increase the apparent density-interval probability by the inverse acceptance probability. The analysis applies the adversary-favorable correction

$$\mathcal{T}_{p,\text{bd}}(l_1, l_0) = \max\{0, \mathcal{T}_p(l_1, l_0) - \lambda_{\text{bd}}^{(w_r)}\},$$

where $\lambda_{\text{bd}}^{(w_r)} \leq -\log_2(1 - U_{\text{clump}})$. The corresponding query bound is

$$q_{\text{bd}}(l_1, l_0) = \min\left\{2^{80}, \left\lfloor 2^{T_{\text{ISD}} - \mathcal{T}_{p,\text{bd}}(l_1, l_0)} \right\rfloor\right\}.$$

The DFR is then evaluated by the same PEB-BSC procedure as above, and the post-rejection CDFR is

$$\text{CDFR}_{\text{bd}} = \max_{(l_1, l_0)} \left\{1 - (1 - \text{DFR}(l_1, l_0))^{q_{\text{bd}}(l_1, l_0)}\right\}.$$

Table 8 gives the resulting post-rejection values used in Table 6.

Table 8: Post-rejection cumulative decryption-failure probability for the specified TriQ-KEM parameter sets under the feasible Guo density-interval attack domain and the bounded-query setting

Parameter set	(L_{bd}, γ)	(l_1, l_0)	$\mathcal{T}_{p,\text{bd}}(l_1, l_0)$	$\text{DFR}(l_1, l_0)$	CDFR_{bd}
TriQ-KEM-128	(384, 7)	(13, 7)	32.09	$\leq 2^{-114.32}$	$\leq 2^{-34.32}$
TriQ-KEM-256	(768, 9)	(17, 9)	48.32	$\leq 2^{-165.63}$	$\leq 2^{-85.63}$
TriQ-KEM-384	(896, 9)	(17, 9)	50.70	$\leq 2^{-204.93}$	$\leq 2^{-124.93}$
TriQ-KEM-512	(1152, 9)	(17, 9)	52.46	$\leq 2^{-226.56}$	$\leq 2^{-146.56}$

The maximizing pairs in Table 8 show that the bounded-density rule restricts the attacker to short dense intervals.

The quantum analysis follows the same evaluation procedure, except that Grover acceleration is only applied to the weak-ciphertext precomputation stage. Thus we set

$$\mathcal{T}_p^Q(l_1, l_0) = \frac{1}{2} \mathcal{T}_p(l_1, l_0),$$

and the admissible number of decryption queries becomes

$$q^Q(l_1, l_0) = \min\left\{2^{80}, \left\lfloor 2^{T_{\text{ISD}}^Q - \mathcal{T}_p^Q(l_1, l_0)} \right\rfloor\right\}.$$

The feasible density-interval parameters (l_1, l_0) are then searched under this quantum budget. The maximizing pairs are reported in Table 9.

Table 9: Pre-rejection reference CDFR for the specified TriQ-KEM parameter sets under the density-interval attack [2] with Grover acceleration in the precomputation stage

Parameter set	$\underline{T}_{\text{ISD}}^Q$	(l_1, l_0)	$\mathcal{T}_p^Q(l_1, l_0)$	DFR(l_1, l_0)	CDFR
TriQ-KEM-128	95.18	(13, 7)	16.07	$\leq 2^{-114.32}$	$\leq 2^{-35.64}$
TriQ-KEM-256	163.49	(53, 27)	85.65	$\leq 2^{-121.31}$	$\leq 2^{-43.48}$
TriQ-KEM-384	229.43	(87, 44)	151.43	$\leq 2^{-128.61}$	$\leq 2^{-50.62}$
TriQ-KEM-512	294.84	(119, 60)	216.95	$\leq 2^{-125.32}$	$\leq 2^{-47.43}$

The bounded-density rejection rule applies identically in the quantum setting. The corresponding post-rejection quantum values are shown in Table 10.

Table 10: Post-rejection cumulative decryption-failure probability for the specified TriQ-KEM parameter sets under the feasible density-interval attack domain, the bounded-query setting, and Grover acceleration in the precomputation stage

Parameter set	(L_{bd}, γ)	(l_1, l_0)	$\mathcal{T}_{p,\text{bd}}^Q(l_1, l_0)$	DFR(l_1, l_0)	CDFR $_{\text{bd}}^Q$
TriQ-KEM-128	(384, 7)	(13, 7)	16.01	$\leq 2^{-114.32}$	$\leq 2^{-35.58}$
TriQ-KEM-256	(768, 9)	(17, 9)	24.15	$\leq 2^{-165.63}$	$\leq 2^{-85.63}$
TriQ-KEM-384	(896, 9)	(17, 9)	25.34	$\leq 2^{-204.93}$	$\leq 2^{-124.93}$
TriQ-KEM-512	(1152, 9)	(17, 9)	26.21	$\leq 2^{-226.56}$	$\leq 2^{-146.56}$

For all four specified TriQ-KEM parameter sets, the post-rejection quantum CDFR bounds do not exceed the corresponding classical post-rejection bounds. Therefore the submitted CDFR thresholds are satisfied in both the classical and quantum settings.

V-C CW-QCSD and ISD Estimates

The public key induces a 2-CW-QCSD instance with target vector $(\mathbf{x}, \mathbf{y})$. The ciphertext randomness induces a 3-CW-QCSD instance with target vector $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$; the bounded-density accepted-set condition on $\mathbf{r}_1$ and $\mathbf{r}_2$ is accounted for below through the entropy-loss correction in Eq. (14). TriQ-KEM keeps the four sparse vectors $(\mathbf{x}, \mathbf{y}, \mathbf{r}_1, \mathbf{r}_2)$ at the same weight and assigns the larger encryption-randomness weight to $\mathbf{e}$:

$$w_{\mathbf{x}} = w_{\mathbf{y}} = w_{\mathbf{r}} < w_{\mathbf{e}}, \quad \mathbf{r}_1, \mathbf{r}_2 \in \mathcal{R}_{w_{\mathbf{r}}}. \quad (11)$$

The ISD estimate used for the parameter-set evaluation is

$$\min \left\{ \underline{T}_{\text{ISD}}(2n, n, (\mathbf{x}, \mathbf{y})), \underline{T}_{\text{ISD}}(3n, n, (\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})) \right\}. \quad (12)$$

This subsection evaluates the induced BSC effect of the weight rule and the corresponding classical and quantum ISD estimates.

V-C.1 Error Distribution and Induced BSC

The induced error vector is

$$\mathbf{e}' = \mathbf{x}\mathbf{r}_2 + \mathbf{r}_1\mathbf{y} + \mathbf{e} \in \mathbb{F}_2^n,$$

which is associated with BSC error probability $p = \mathbb{P}[e'_k = 1]$ for any fixed coordinate k by Lemma V.3. The following approximate statement captures the effect of moving encryption-randomness weight from $\mathbf{r}_1, \mathbf{r}_2$ to $\mathbf{e}$, with $\mathbf{x}, \mathbf{y}$ fixed, under the coordinate-wise BSC model used in the DFR analysis.

Theorem V.7 (BSC-approximation effect of weight redistribution). *Let $n \gg w_0 > w_{\mathbf{x}}$ be three positive integers, and let $n > w_{\mathbf{x}}^2$. For $0 \leq \delta \leq w_0 - w_{\mathbf{x}}$, define the weights*

$$w_{\mathbf{x}} = w_{\mathbf{y}} \geq 67, \quad w_{\mathbf{r}} = w_0 - \delta, \quad w_{\mathbf{e}} = w_0 + 2\delta,$$

where $w_{\mathbf{r}}$ is the common weight used for $\mathbf{r}_1$ and $\mathbf{r}_2$. Then, in the BSC approximation, the coordinate error probability $p_{\text{BSC}}(\delta)$ decreases as δ increases.

Proof location. See Appendix A-C.

Theorem V.7 gives the BSC approximation showing that an unbalanced weight distribution within $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$ reduces the approximated induced BSC error probability. This motivates Eq. (11) from the decryption-failure side; the ISD estimates are evaluated below.

In addition to weight redistribution, the length n affects the induced BSC. The selected values of n are primitive primes above the RSRM codeword length $n_1 n_2$.

Theorem V.8. *Let the weights $w_{\mathbf{x}}, w_{\mathbf{y}}, w_{\mathbf{r}}, w_{\mathbf{e}}$ be fixed positive integers, where $w_{\mathbf{r}}$ is the common weight used for $\mathbf{r}_1$ and $\mathbf{r}_2$. For the parameter regime considered in this document, assume the sufficient inequalities used in the proof hold, in particular $n > AB - 1$ for the relevant convolution-weight pairs (A, B) and $p_U, p_V, p_W < 1/2$. Define $P(n) = \mathbb{P}[e'_k = 1]$ with the probability given by Lemma V.3. Then $P(n)$ is strictly decreasing in n over this regime.*

Proof location. See Appendix A-C.

Theorem V.8 states that, within the TriQ parameter regime and with all weights fixed, increasing n lowers the induced BSC error probability and hence lowers the DFR. The truncation by $\ell = n - n_1 n_2$ coordinates is included in the concrete CW-QCSD and ISD estimates.

V-C.2 Structured Errors and ISD Complexity

The same weight distribution also affects the best known CW-QCSD-based ISD attacks. ISD algorithms are the best known approach to attack syndrome-decoding instances in this setting and are the tool used here to evaluate concrete bit-complexity. The estimates include Prange [7], Stern [8], Both-May [9], May-Ozerov [10], and BJMM [11], together with quantum Prange for the quantum bit-complexity estimates.

Structured ISD Estimate. For the unbalanced error distribution used in TriQ-KEM, the ISD permutation success probability is derived from a structured combinatorial model; the detailed derivation, the asymptotic optimal-allocation theorem (Theorem B.1), and the oracle-complexity analysis are provided in Appendix B. The overall ISD bit-complexity for a structured error $\mathbf{e}$ is estimated as

$$\underline{T}_{\text{ISD}}(n, k, \mathbf{e}) = \min\{T_{\text{ISD}}, T'_{\text{ISD}}\} - \log_2 \sqrt{\#\text{targets}}, \quad (13)$$

where T_{ISD} and T'_{ISD} are respectively the unstructured and structured ISD bit-complexities, and $\#\text{targets} = n$ under the DOOM attack [12]. The resulting estimates for the specified TriQ-KEM parameter sets are reported in Tables 11 and 12.

V-C.3 Concrete ISD Evaluation

Table 11 reports the bit-complexity estimates for the 2-CW-QCSD instance $\underline{T}_{\text{ISD}}(2n, n, (\mathbf{x}, \mathbf{y}))$. Table 12 reports the bit-complexity estimates for the 3-CW-QCSD instance $\underline{T}_{\text{ISD}}(3n, n, (\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}))$. For direct comparison with the standardized HQC parameter sets, the tables include HQC-128 and HQC-256 next to TriQ-KEM-128 and TriQ-KEM-256. For the TriQ-KEM rows, boldface marks the values used in the security summary: the final classical ISD entries come from the 3-CW-QCSD table, while the Q-Prange minima come from the 2-CW-QCSD table.

Table 11: Concrete ISD bit-complexities for the 2-CW-QCSD instance $\underline{T}_{\text{ISD}}(2n, n, (\mathbf{x}, \mathbf{y}))$.

Parameter set	Prange	Stern	BothMay	MayOzerov	BJMM	Q-Prange
HQC-128	166.33	145.31	145.44	145.44	145.63	94.39
TriQ-KEM-128	168.09	146.97	147.18	147.18	147.32	95.18
HQC-256	300.46	275.60	275.76	275.76	275.82	162.84
TriQ-KEM-256	302.07	277.14	277.08	277.09	277.40	163.49
TriQ-KEM-384	432.42	405.27	404.62	404.63	404.97	229.43
TriQ-KEM-512	562.11	533.39	532.44	532.44	533.16	294.84

Table 12: Concrete ISD bit-complexities for the 3-CW-QCSD instance $\underline{T}_{\text{ISD}}(3n, n, (\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}))$.

Parameter set	Prange	Stern	BothMay	MayOzerov	BJMM	Q-Prange
HQC-128	167.98	147.25	147.33	147.33	147.52	96.70
TriQ-KEM-128	167.75	146.67	146.80	146.80	146.98	96.49
HQC-256	301.99	277.36	277.46	277.46	277.55	165.08
TriQ-KEM-256	302.81	277.56	277.04	277.04	277.35	165.33
TriQ-KEM-384	432.76	405.23	404.13	404.13	404.46	231.07
TriQ-KEM-512	562.71	533.52	532.06	532.06	532.77	296.60

For the specified TriQ-KEM parameter sets, the final classical ISD estimates are determined by the 3-CW-QCSD entries, namely 146.67, 277.04, 404.13, and 532.06 bits. The Q-Prange bit-complexity estimates are determined by the smaller value over the 2-CW-QCSD and 3-CW-QCSD entries; for all four specified instances, these minima come from the 2-CW-QCSD entries and are 95.18, 163.49, 229.43, and 294.84 bits.

The bounded-density rejection rule reduces the sampling space for $\mathbf{r}_1$ and $\mathbf{r}_2$. It does not restrict $\mathbf{x}$, $\mathbf{y}$, or $\mathbf{e}$. Let U_{clump} be the rejection-probability upper bound from Lemma III.1 and set

$$\lambda_{\text{bd}}^{(w_{\mathbf{r}})} = -\log_2(1 - U_{\text{clump}}).$$

The resulting first-order correction for the bounded-density-conditioned 3-CW-QCSD estimate is bounded by

$$T_{\text{ISD},\text{bd}} \geq T_{\text{ISD},\text{base}} - 2\lambda_{\text{bd}}^{(w_{\mathbf{r}})}, \quad (14)$$

where the factor 2 accounts only for $\mathbf{r}_1$ and $\mathbf{r}_2$. For the rejection parameters in Table 4, the per-instance values of $\lambda_{\text{bd}}^{(w_{\mathbf{r}})}$ are at most 0.0613, giving $2\lambda_{\text{bd}}^{(w_{\mathbf{r}})} \leq 0.123$ bits. Thus the bounded-density correction changes the displayed 3-CW-QCSD ISD estimates by at most 0.123 bits.

The 2-CW-QCSD and 3-CW-QCSD minima remain close for all four TriQ-KEM parameter sets. Thus the weight rule in Eq. (11) lowers the induced BSC error probability while maintaining the stated classical and quantum ISD estimates for both induced CW-QCSD instances.

VI Implementation and Performance

The code includes a portable reference implementation, an AVX2 optimized implementation, and known-answer test vectors under `Test_Vectors/`.

VI-A Implementation Overview

The implementations follow the TriQ-KEM algorithms specified in Section III. The submitted implementations are written in C and compiled as C99 code. The reference implementation is

portable and does not rely on platform-specific instruction sets. The optimized implementation targets mainstream 64-bit x86 platforms with AVX2/PCLMUL/BMI support. Both implementations use the API_PKC KEM interface. The hash and XOF calls, including deterministic expansion from seeds and encryption randomness, use the API_PKC auxiliary functions; KAT seeds are generated by the submitted deterministic DRNG harness.

The main software costs are as follows:

- **Key generation** is dominated by fixed-weight sampling of $\mathbf{x}, \mathbf{y}$, generation of $\mathbf{h}$, and the product $\mathbf{h}\mathbf{y}$;
- **Encapsulation** is dominated by sampling $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$, the products $\mathbf{h}\mathbf{r}_2$ and $\mathbf{s}\mathbf{r}_2$, and the encoding of the RSRM plaintext block;
- **Decapsulation** is dominated by the product $\mathbf{u}\mathbf{y}$, the BM-FHT decoding of the RSRM code, and the reencryption check used by the FO transform.

The bounded-density sampler for $\mathbf{r}_1$ and $\mathbf{r}_2$ is a public rejection step: after a fixed-weight candidate is generated, the implementation scans the prescribed cyclic windows and rejects candidates that violate the public bound. The public rejection test is implemented as a fixed-window scan in constant time with respect to secret data. Decapsulation recomputes the candidate ciphertext and follows the FO implicit-rejection path without data-dependent early exit on secret values.

VI-B Performance Evaluation

Tables 13 and 14 report the measured CPU-cycle counts for the reference (ref) and optimized (AVX2) implementations of the four TriQ-KEM instances. The comparison uses the listed HQC-128, HQC-256, HQC-384, and HQC-512 implementations. The HQC parameter sets and raw cycle counts are collected in Appendix C-A.

Benchmarking methodology. Each operation (KeyGen, Encaps, Decaps) is executed 10 000 times. Cycle counts are measured via RDTSC and reported as per-call median values. The benchmark platform is an Intel® Core™ i7-10700 CPU @ 2.90 GHz (8 cores, 16 threads) with 32 GB DDR4-2933 MHz RAM, running Microsoft Windows 11 Pro (24H2, 64-bit). The build environment is MSYS2 UCRT64 with GCC 15.2.0 (x86_64-w64-mingw32). All builds use `-std=c99 -O3`; optimized builds additionally use `-mavx -mavx2 -mbmi -mpclmul`.

Polynomial multiplication architecture. The dominant cost in both TriQ-KEM and HQC is polynomial multiplication in $\mathcal{R} = \mathbb{F}_2[X]/(X^n - 1)$. The reference implementation uses scalar word operations with schoolbook base multiplication and recursive Karatsuba for multiplication in $\mathcal{R}$. The optimized implementation uses AVX2/PCLMUL-enabled kernels, with instance-specific Karatsuba or Toom-Cook-style decomposition, for the same ring operations. The bounded-density rejection sampler for $\mathbf{r}_1, \mathbf{r}_2$ is the additional sampling step in the TriQ-KEM encryption path. For each bounded-density sampling call, the expected number of candidate draws is at most $1/(1 - U_{\text{clump}}) \leq 1.044$ for all four instances.

Reference-implementation comparison. Table 13 compares the TriQ-KEM and HQC reference implementations. The reduction grows from 9.6% at the 128-bit level to 30.2% at the 384-bit level, reflecting the cumulative effect of the smaller ring length n and the shorter RSRM code length $n_1 n_2$ in the TriQ-KEM parameter sets.

Table 13: Reference-implementation runtime comparison (median cycles per call, 10 000 repetitions)

Category	KeyGen	Encaps	Decaps	Total cycles	Reduction
128-bit Security Level					
HQC-128	4 367 571	8 419 044	12 731 609	25 518 224	–
TriQ-KEM-128	3 964 805	7 609 049	11 485 306	23 059 160	9.6%
HQC-256	27 303 132	54 149 813	81 624 237	163 077 182	–
TriQ-KEM-256	22 098 875	43 874 667	65 778 727	131 752 269	19.2%
HQC-384	84 072 416	167 482 187	252 261 030	503 815 633	–
TriQ-KEM-384	58 835 996	116 972 259	175 925 165	351 733 420	30.2%
HQC-512	151 314 232	299 869 152	450 564 790	901 748 174	–
TriQ-KEM-512	107 949 010	214 011 770	320 423 746	642 384 526	28.8%

Optimized-implementation performance. Table 14 reports the measured cycle counts for the TriQ-KEM AVX2-optimized implementations. All four instances use AVX2-oriented low-level arithmetic and vector routines validated on the benchmark platform.

Table 14: Optimized-implementation runtime (median cycles per call, 10 000 repetitions)

Instance	KeyGen	Encaps	Decaps	Total cycles
TriQ-KEM-128	370 080	405 937	527 445	1 303 462
TriQ-KEM-256	852 679	1 353 561	1 929 149	4 135 389
TriQ-KEM-384	1 844 526	2 896 869	4 433 519	9 174 914
TriQ-KEM-512	5 010 708	7 960 565	11 200 361	24 171 634

VI-C Known-Answer Tests

Known-answer tests for TriQ-KEM follow the API_PKC KEM record format. The files are placed directly under `Test_Vectors/`, with one file per submitted parameter set. For each parameter set, every deterministic KAT record contains the fields

Count, Seed_Len, Seed, PK_Len, PK, SK_Len, SK,
CT_Len, CT, SS_Len, SS.

VI-D Feature Statement

TriQ-KEM provides the following algorithmic characteristics.

- **Innovation.** TriQ-KEM combines unbalanced PKE encryption-vector weights with a public bounded-density rejection rule. The parameter evaluation jointly accounts for natural DFR, attack-aware CDFR, CW-QCSD-based ISD estimates, and quantum-Prange bit-complexity estimates.
- **Simplicity.** TriQ-KEM is obtained by applying the Fujisaki–Okamoto transform to TriQ-PKE. The construction uses the TriQ-PKE key generation, encryption, and decryption algorithms directly and does not introduce an additional code-based primitive.

- **Flexibility.** The same specification supports TriQ-KEM-128, TriQ-KEM-256, TriQ-KEM-384, and TriQ-KEM-512. The serialized public keys, secret keys, ciphertexts, and shared secrets follow a consistent format across all submitted parameter sets.
- **Compatibility.** TriQ-KEM is designed to pair with TriQ-KEX as a code-based KEM/KEX family with matching security categories, common implementation interfaces, and compatible API_PKC auxiliary-function usage for KAT generation and benchmarking.
- **Scalability.** The parameter-selection method scales the ring length, RSRM code parameters, vector weights, and security estimates across the four submitted security levels.
- **Implementation performance.** The submission includes both reference and optimized implementations. The measured key-generation, encapsulation, and decapsulation cycle counts are reported in Section [VI-B](#).

VI-E Intellectual Property Notice

The following patents may cover the submitted TriQ-KEM algorithm:

- US9094189 B2, *Cryptographic method for communicating confidential information*;
- FR 10/51190, *Procédé cryptographique de communication d'une information confidentielle*.

A Proofs for Failure Bounds

This section gives the proof details used by the failure and concrete-security analysis in the main text. The order follows the dependency chain used in the main body: first the BSC and bounded-density rejection arguments, then the RM/RSRM failure bounds, and then the weight-redistribution results.

A-A BSC and Bounded-Density Rejection Proofs

Proof of Lemma V.2. According to Proposition 6.1.1. in [1],

$$\Pr[z_k = 1] = \sum_{\substack{1 \leq \ell \leq \min(w_{\mathbf{x}}, w_{\mathbf{r}}) \\ \ell \text{ is odd}}} \frac{\binom{n}{\ell} \binom{n-\ell}{w_{\mathbf{x}}-\ell} \binom{n-w_{\mathbf{x}}}{w_{\mathbf{r}}-\ell}}{\binom{n}{w_{\mathbf{x}}} \binom{n}{w_{\mathbf{r}}}}.$$

Using the identity $\binom{n}{\ell} \binom{n-\ell}{w_{\mathbf{x}}-\ell} = \binom{n}{w_{\mathbf{x}}} \binom{w_{\mathbf{x}}}{\ell}$, we obtain $\frac{\binom{n}{\ell} \binom{n-\ell}{w_{\mathbf{x}}-\ell}}{\binom{n}{w_{\mathbf{x}}}} = \binom{w_{\mathbf{x}}}{\ell}$. Thus, we get Eq. (5). $\square$

Proof of Lemma V.3. Fix an index k . By Lemma V.2 and the given distributions, the three random variables

$$U = (\mathbf{x} \cdot \mathbf{r}_2)_k, \quad V = (\mathbf{r}_1 \cdot \mathbf{y})_k, \quad W = e_k$$

are mutually independent Bernoulli random variables with parameters p_U , p_V , and p_W , respectively. Over $\mathbb{F}_2$, subtraction equals addition, so $e'_k = U + V + W$.

Using the mutual independence of U, V, W ,

$$\begin{aligned} \Pr[e'_k = 1] &= \Pr(U = 1, V = 0, W = 0) + \Pr(U = 0, V = 1, W = 0) \\ &\quad + \Pr(U = 0, V = 0, W = 1) + \Pr(U = 1, V = 1, W = 1) \\ &= p_U(1 - p_V)(1 - p_W) + (1 - p_U)p_V(1 - p_W) \\ &\quad + (1 - p_U)(1 - p_V)p_W + p_U p_V p_W \\ &= p_U + p_V + p_W - 2p_U p_V - 2p_U p_W - 2p_V p_W + 4p_U p_V p_W. \end{aligned}$$

This completes the proof. $\square$

Proof of Lemma III.1. The trivial bound $p_{\text{rej}}^{(w)} \leq 1$ is always available, which is the first term inside the minimum in Eq. (2). It therefore remains to establish the second term under the assumption $\gamma n > w(L - 1)$.

Choose a support S uniformly among all w -subsets of $\mathbb{Z}_n$, then choose one point of S as a root and list the support points in cyclic order. Let $G_1, \dots, G_w$ be the positive cyclic gaps between consecutive support points, with indices taken modulo w . The rooted-support representation is a bijection between rooted supports and pairs consisting of a root position and a positive composition

$$G_1 + \dots + G_w = n.$$

Therefore, conditioned on the root, the gap vector is uniform over the $\binom{n-1}{w-1}$ positive compositions of n into w parts.

By Eq. (1), a support is rejected iff some length- L cyclic window contains at least $\gamma + 1$ support points. Equivalently, for some cyclic index i , the first-to-last distance of $\gamma + 1$ consecutive support points is at most $L - 1$. Define

$$E_i = \{G_i + G_{i+1} + \dots + G_{i+\gamma-1} \leq L - 1\}.$$

Then $\{S \notin A_{L,\gamma}^{(w)}\} = \bigcup_i E_i$.

Let $F_i = E_i \cap E_{i-1}^c$. If all E_i occurred simultaneously, summing their defining inequalities over the w cyclic indices would count every gap exactly γ times, giving

$$\gamma n = \gamma \sum_{i=1}^w G_i \leq w(L-1),$$

which contradicts the assumption $\gamma n > w(L-1)$. Hence the E_i cannot all occur, so whenever the rejection event occurs the nonempty cyclic set $\{i : E_i \text{ occurs}\}$ has at least one start point i at which E_i occurs but E_{i-1} does not, i.e., F_i occurs. Therefore

$$\mathbf{1}_{\bigcup_i E_i} \leq \sum_i \mathbf{1}_{F_i}, \quad \text{so} \quad p_{\text{rej}}^{(w)} \leq w \Pr[F_1]$$

by cyclic symmetry.

It remains to count $\Pr[F_1]$. Write

$$A = G_1 + \cdots + G_{\gamma-1}, \quad B = G_\gamma, \quad C = G_0.$$

Then E_1 is the event $A + B \leq L-1$, while E_0^c is the event $A + C \geq L$. Fix $A = a$. The shared $\gamma-1$ positive gaps summing to a can be chosen in $\binom{a-1}{\gamma-2}$ ways, and a ranges from $\gamma-1$ to $L-2$. For a fixed a , the gap $B = b$ ranges from 1 to $L-1-a$, and $C = c$ satisfies $c \geq L-a$; the remaining $w-\gamma-1$ gaps are positive and sum to $n-a-b-c$, contributing $\binom{n-a-b-c-1}{w-\gamma-2}$ compositions. Summing over c and then over b with the hockey-stick identity gives

$$\sum_{b=1}^{L-1-a} \sum_{c=L-a}^{\infty} \binom{n-a-b-c-1}{w-\gamma-2} = \binom{n-L}{w-\gamma} - \binom{n-2L+1+a}{w-\gamma}.$$

Dividing by the total number $\binom{n-1}{w-1}$ of gap compositions and multiplying by w yields

$$p_{\text{rej}}^{(w)} \leq \frac{w}{\binom{n-1}{w-1}} \sum_{a=\gamma-1}^{L-2} \binom{a-1}{\gamma-2} \left(\binom{n-L}{w-\gamma} - \binom{n-2L+1+a}{w-\gamma} \right),$$

which is the second term in Eq. (2). Combining with the trivial bound completes the proof. $\square$

A-B Proofs for RM and RSRM Failure Bounds

A-B.1 Proof of Theorem V.4

The proof starts from the algebraic structure of the relevant RM codes. According to [13], all non-zero vectors in $\mathbb{F}_2^k$ constitute a generator matrix $\mathbf{S}_k$ for a k -dimensional Simplex code with parameters $[2^k-1, k, 2^{k-1}]_2$. The Simplex code is the unique constant-weight linear code. That is, any single-weight linear code is either a Simplex code or a repetition (possibly with additional zero bits) of a Simplex code. A generator matrix of an RM code C with parameters $[128s, 8, 64s]_2$ can be expressed as:

$$\mathbf{G}_{\text{RM}} = \begin{pmatrix} \mathbf{G}'_{\text{RM}} \\ \mathbf{1} \end{pmatrix} = \begin{pmatrix} s \cdot \mathbf{S}_7 & \mathbf{0}_{7 \times s} \\ \mathbf{1} & \mathbf{1} \end{pmatrix},$$

where $\mathbf{S}_7$ denotes a generator matrix of the 7-dimensional Simplex code C_s . This means $\mathbf{G}'_{\text{RM}}$ contains all vectors of $\mathbb{F}_2^7$, each repeated s times, i.e., $\mathbf{G}'_{\text{RM}}$ generates a repeated Simplex code C'_s with s zero coordinates. The RM code C has 255 nonzero codewords, among which 254 have weight $64s$ and one has weight $128s$. Here, we partition the codewords of weight $64s$ in the RM code C into two distinct sets. The first set is denoted as $\mathcal{S}_1$, which contains all nonzero

codewords of C'_s . The second set is defined as $\mathcal{S}_2 = \{\mathbf{1}_{128s} + \mathbf{c} \mid \mathbf{c} \in \mathcal{S}_1\}$. Thus, the code C can be expressed as the following union:

$$C = \mathcal{S}_1 \cup \mathcal{S}_2 \cup \{\mathbf{1}_{128s}, \mathbf{0}_{128s}\}.$$

Considering the relative relationship between errors and codewords of the RM code, the following lemma holds.

Lemma A.1. *Let $\mathbf{c}_1$ and $\mathbf{c}_2$ be two nonzero codewords in the $[128s, 8, 64s]_2$ RM code C , satisfying $|\text{Supp}(\mathbf{c}_1) \cup \text{Supp}(\mathbf{c}_2)| = 128s$. Then, there does not exist an error vector $\mathbf{e} \in \mathbb{F}_2^{128s}$ of weight less than $64s$ such that $d(\mathbf{c}_1, \mathbf{e}) = d(\mathbf{c}_2, \mathbf{e}) = \text{wt}(\mathbf{e})$ or $d(\mathbf{c}_1, \mathbf{e}), d(\mathbf{c}_2, \mathbf{e}) < \text{wt}(\mathbf{e})$.*

Proof. Since $|\text{Supp}(\mathbf{c}_1) \cup \text{Supp}(\mathbf{c}_2)| = 128s$, and the codewords of the RM code have only three possible weights, namely 0 , $64s$, and $128s$, the relationship among the supports of $\mathbf{c}_1$, $\mathbf{c}_2$, and $\mathbf{e}$ can only take one of the following two forms:

$$\begin{pmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{64s} & \mathbf{0}_{64s} \\ \mathbf{0}_{64s} & \mathbf{1}_{64s} \\ \mathbf{e}_1 & \mathbf{e}_2 \end{pmatrix} \text{ or } \begin{pmatrix} \mathbf{1}_{64s} & \mathbf{0}_{64s} \\ \mathbf{1}_{64s} & \mathbf{1}_{64s} \\ \mathbf{e}_1 & \mathbf{e}_2 \end{pmatrix}.$$

For the first case, we have $d(\mathbf{c}_1, \mathbf{e}) = \text{wt}(\mathbf{c}_1) - \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_2)$ and $d(\mathbf{c}_2, \mathbf{e}) = \text{wt}(\mathbf{c}_2) + \text{wt}(\mathbf{e}_1) - \text{wt}(\mathbf{e}_2)$. It follows that if $d(\mathbf{c}_1, \mathbf{e}) = d(\mathbf{c}_2, \mathbf{e}) = \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_2)$, then $\text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_2) = \frac{\text{wt}(\mathbf{c}_1) + \text{wt}(\mathbf{c}_2)}{2} = 64s$. This contradicts the assumption. If $d(\mathbf{c}_1, \mathbf{e}), d(\mathbf{c}_2, \mathbf{e}) < \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_2)$, then $\text{wt}(\mathbf{e}_1), \text{wt}(\mathbf{e}_2) > 32s$. This also leads to a contradiction.

Using a similar method, the latter case can also be proven impossible. $\square$

Based on Lemma A.1 and the structure of the codeword space of the RM code, the following lemma can be established.

Lemma A.2. *Let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m$ be m ($m \geq 3$) distinct nonzero codewords in the $[128s, 8, 64s]_2$ RM code C . Let $\mathbf{e}$ be an error vector in $\mathbb{F}_2^{128s}$ of weight $w \leq 64s - 1$. Then the following results hold.*

- (1) *If the error vector $\mathbf{e}$ satisfies $d(\mathbf{c}_1, \mathbf{e}) = \dots = d(\mathbf{c}_m, \mathbf{e}) = w$, then $w \geq 40s$.*
- (2) *If the error vector $\mathbf{e}$ satisfies $d(\mathbf{c}_1, \mathbf{e}), \dots, d(\mathbf{c}_m, \mathbf{e}) < w$, then $w \geq 40s + 2$.*

Proof. Let $\mathbf{G}_m = \begin{pmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_m \end{pmatrix}$, and let C_m denote the linear code generated by $\mathbf{G}_m$. According to

Lemma A.1, the codewords $\mathbf{c}_1, \dots, \mathbf{c}_m$ contain no codeword of weight $128s$, and the sum of any two such codewords does not have weight $128s$. It follows that C_m contains only nonzero codewords of weight $64s$. According to [13], only Simplex codes are single-weight codes. Therefore, C_m is a Simplex code of dimension $\text{Rank}(\mathbf{G}_m)$ with zero coordinates.

Now consider the following cases separately. If $\text{Rank}(\mathbf{G}_m) = m$, then $\mathbf{G}_m$ is an $m \times 128s$ full-rank matrix composed of all vectors of $\mathbb{F}_2^m$, each repeated 2^{7-m} times; thus, $\mathbf{G}_m$ contains $s \cdot 2^{7-m} \cdot \binom{m}{\mu}$ columns of weight μ .

Suppose the support of the error vector $\mathbf{e}$ corresponds to λ_μ columns of weight μ in $\mathbf{G}_m$, i.e.,

$$\text{wt}(\mathbf{e}) = \sum_{\mu=0}^m \lambda_\mu,$$

where $1 \leq \lambda_\mu \leq s \cdot 2^{7-m} \cdot \binom{m}{\mu}$. Furthermore, from

$$d(\mathbf{c}_i, \mathbf{e}) = \text{wt}(\mathbf{c}_i) + \text{wt}(\mathbf{e}) - 2|\text{Supp}(\mathbf{c}_i) \cap \text{Supp}(\mathbf{e})|,$$

it follows that for $d(\mathbf{c}_i, \mathbf{e}) = \text{wt}(\mathbf{e})$, we have $|\text{Supp}(\mathbf{c}_i) \cap \text{Supp}(\mathbf{e})| = 32s$; and for $d(\mathbf{c}_i, \mathbf{e}) < \text{wt}(\mathbf{e})$, we have $|\text{Supp}(\mathbf{c}_i) \cap \text{Supp}(\mathbf{e})| \geq 32s + 1$. Consequently, if condition (1) or (2) is satisfied, then

$$\sum_{\mu=1}^m \lambda_{\mu} \cdot \mu = \sum_{i=1}^m \sum_{j=1}^{128s} e_j \cdot c_{i,j} \geq 32s \cdot m \quad \text{or} \quad (32s + 1) \cdot m.$$

The Hamming weight of the error vector $\mathbf{e}$ is minimized when the weighted sum of the columns corresponding to the support of $\mathbf{e}$ is maximized. The core strategy is to maximize the “weighted contribution per unit λ_{μ} ,” since a larger μ means each unit of λ_{μ} contributes more to the weighted sum. Therefore, we prioritize using λ_{μ} with the largest μ (up to its upper bound) until the weighted sum constraint is met. Following this approach, the minimum weight $\text{wt}(\mathbf{e})$ for different values of m can be determined as follows:

m	3	4	5	6	7
$\text{wt}(\mathbf{e})$ under condition (1)	$\geq 40s$	$\geq 40s$	$\geq 44s$	$\geq 44s$	$\geq \lceil \frac{93s}{2} \rceil$
$\text{wt}(\mathbf{e})$ under condition (2)	$\geq 40s + 2$	$\geq 40s + 2$	$\geq 44s + 2$	$\geq 44s + 2$	$\geq \lfloor \frac{93s+5}{2} \rfloor$

Thus, for $\text{Rank}(\mathbf{G}_m) = m \geq 3$, (1) and (2) hold.

Next, consider the other case: $\text{Rank}(\mathbf{G}_m) = m' < m$. Since $m \geq 3$, we have $m' \geq 2$. When $m = 3$ and $m' = 2$, without loss of generality, after appropriate permutation, the intersection relationship among the supports of $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$, and $\mathbf{e}$ can be expressed in the following form:

$$\begin{pmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \mathbf{c}_3 = \mathbf{c}_1 + \mathbf{c}_2 \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{32s} & \mathbf{1}_{32s} & \mathbf{0}_{32s} & \mathbf{0}_{32s} \\ \mathbf{0}_{32s} & \mathbf{1}_{32s} & \mathbf{1}_{32s} & \mathbf{0}_{32s} \\ \mathbf{1}_{32s} & \mathbf{0}_{32s} & \mathbf{1}_{32s} & \mathbf{0}_{32s} \\ \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 & \mathbf{e}_4 \end{pmatrix}.$$

If

$$d(\mathbf{c}_1, \mathbf{e}) = d(\mathbf{c}_2, \mathbf{e}) = d(\mathbf{c}_3, \mathbf{e}) = w$$

or

$$d(\mathbf{c}_1, \mathbf{e}), d(\mathbf{c}_2, \mathbf{e}), d(\mathbf{c}_3, \mathbf{e}) < w,$$

then we have

$$\begin{cases} \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_2) = 32s \\ \text{wt}(\mathbf{e}_2) + \text{wt}(\mathbf{e}_3) = 32s \\ \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_3) = 32s \end{cases} \quad \text{or} \quad \begin{cases} \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_2) > 32s \\ \text{wt}(\mathbf{e}_2) + \text{wt}(\mathbf{e}_3) > 32s \\ \text{wt}(\mathbf{e}_1) + \text{wt}(\mathbf{e}_3) > 32s \end{cases}.$$

This leads to $\text{wt}(\mathbf{e}) = \sum_{i=1}^4 \text{wt}(\mathbf{e}_i) \geq 48s$ or $48s + 2$.

When $m \geq 4$ and $m' < m$, since $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m$ are distinct nonzero codewords, we have $m' \geq 3$. Without loss of generality, assume

$$\text{Rank} \begin{pmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_{m'} \end{pmatrix} = m' \geq 3.$$

Then, according to the previous proof, any error $\mathbf{e}$ satisfying condition (1) or (2) must satisfy $\text{wt}(\mathbf{e}) \geq 40s$ or $40s + 2$.

In summary, any error vector $\mathbf{e}$ satisfying condition (1) has a Hamming weight of at least $40s$; any error vector $\mathbf{e}$ satisfying condition (2) has a Hamming weight of at least $40s + 2$. $\square$

We now prove Theorem [V.4](#).

Proof of Theorem V.4. Let $\mathbf{c}$ be a codeword of code C , and $\mathbf{y} = \mathbf{c} + \mathbf{e}$ be the received word. Let $S(\mathbf{e})$ denote the set of all codewords closest to $\mathbf{e}$, i.e., for any $\mathbf{c}_1 \in C \setminus S(\mathbf{e})$ and $\mathbf{c}_2 \in S(\mathbf{e})$, we have

$$d(\mathbf{e}, \mathbf{c}_1) > d(\mathbf{e}, \mathbf{c}_2).$$

It follows that

$$d(\mathbf{y} = \mathbf{c} + \mathbf{e}, \mathbf{c} + \mathbf{c}_1) = d(\mathbf{e}, \mathbf{c}_1) > d(\mathbf{c} + \mathbf{e}, \mathbf{c} + \mathbf{c}_2) = d(\mathbf{e}, \mathbf{c}_2),$$

so under ML decoding, $\mathbf{y}$ will be decoded as $\mathbf{c} + \mathbf{c}_2$. If $|S(\mathbf{e})| > 1$, then a codeword is randomly selected from $S(\mathbf{e})$ as the decoding result. Decoding succeeds only if $\mathbf{e}$ is decoded as the zero vector (i.e., $\mathbf{c}_2 = \mathbf{0}$); otherwise, decoding fails. Specifically, the following three cases may occur:

- **Case 1:** If for all nonzero codewords $\mathbf{c}' \in C$, we have $d(\mathbf{0}, \mathbf{e}) < d(\mathbf{e}, \mathbf{c}')$, then $S(\mathbf{e})$ contains only the zero codeword, and ML decoding will definitely succeed.
- **Case 2:** If there exists a nonzero codeword $\mathbf{c}' \in C$ such that $d(\mathbf{0}, \mathbf{e}) > d(\mathbf{e}, \mathbf{c}')$, then $S(\mathbf{e})$ does not contain the zero codeword, and decoding definitely fails.
- **Case 3:** If there exists at least one nonzero codeword $\mathbf{c}' \in C$ such that $d(\mathbf{0}, \mathbf{e}) = d(\mathbf{e}, \mathbf{c}')$, then $S(\mathbf{e})$ contains the zero codeword and $|S(\mathbf{e})| \geq 2$. In this case, the probability of decoding failure is $\frac{|S(\mathbf{e})|-1}{|S(\mathbf{e})|}$.

For a random error $\mathbf{e}$ of Hamming weight w , computing the decoding error probability is equivalent to computing the expected number of error vectors that cause decoding failure, or the expected number of error vectors that are correctly decoded. Let $\mathcal{S}_w$ denote the set of all error vectors in $\mathbb{F}_2^{128s}$ of weight w . Then the probability of correction failure for errors of weight w , $P_{\text{fail}}(C, w)$, is

$$P_{\text{fail}}(C, w) = \frac{\sum_{\mathbf{e} \in \mathcal{S}_w} \delta(\mathbf{e})}{\binom{128s}{w}}, \quad \delta(\mathbf{e}) = \begin{cases} 1, & \text{if } \mathbf{0} \notin S(\mathbf{e}), \\ \frac{|S(\mathbf{e})|-1}{|S(\mathbf{e})|}, & \text{if } \mathbf{0} \in S(\mathbf{e}). \end{cases} \quad (15)$$

Therefore, estimating the weight-wise failure probability $P_{\text{fail}}(C, w)$ for the RM code C is equivalent to counting $S(\mathbf{e})$.

Since the covering radius of a linear code (i.e., the maximum error weight it can correct) does not exceed its minimum distance [14], the error correction capability of an RM code with parameters $[128s, 8, 64s]_2$ is at most $64s$. To simplify the calculation, we only compute the correction probability for errors of weight $w \leq 64s - 1$. Errors of weight $w \geq 64s$ are uniformly considered uncorrectable. Under this setting, there is no error vector $\mathbf{e}$ satisfying $d(\mathbf{e}, \mathbf{1}) \leq d(\mathbf{e}, \mathbf{0})$; therefore, we only need to consider the case where error $\mathbf{e}$ is decoded to a codeword of weight $64s$. Next, we analyze $S(\mathbf{e})$ based on the relative relationship between the support of codewords and the support of error $\mathbf{e}$, and then estimate the number of error vectors that cause decoding failure.

Considering the relationship between a single nonzero codeword $\mathbf{c}_1$ and the error vector $\mathbf{e}$: The RM code C contains 254 codewords of weight $64s$. Without loss of generality, we can represent the support relationship between a codeword $\mathbf{c}_1$ of weight $64s$ and the error vector $\mathbf{e}$ using the following matrix structure:

$$\begin{pmatrix} \mathbf{c}_1 \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{64s} & \mathbf{0}_{64s} \\ \mathbf{e}_1 & \mathbf{e}_2 \end{pmatrix}, \quad (16)$$

where $\text{wt}(\mathbf{e}_1) = u$ and $\text{wt}(\mathbf{e}_2) = w - u$.

The distance between codeword $\mathbf{c}_1$ and error vector $\mathbf{e}$ can be expressed as

$$d(\mathbf{c}_1, \mathbf{e}) = \text{wt}(\mathbf{c}_1) + \text{wt}(\mathbf{e}) - 2|\text{Supp}(\mathbf{c}_1) \cap \text{Supp}(\mathbf{e})| = 64s + w - 2u.$$

If $\mathbf{0} \notin S(\mathbf{e})$, then $d(\mathbf{c}_1, \mathbf{e}) < d(\mathbf{0}, \mathbf{e}) = w$, which implies $u \geq 32s + 1$. In this case, the error vector $\mathbf{e}$ causes decoding failure of Case 2. For a single codeword $\mathbf{c}_1$, the number of error vectors $\mathbf{e}$ of weight $w \leq 64s - 1$ satisfying $u \geq 32s + 1$ is

$$N_{w,1} = \sum_{u=32s+1}^{64s-1} \binom{64s}{u} \binom{64s}{w-u}.$$

Since C has 254 codewords of weight $64s$, the total number of error vectors satisfying this condition is at most $254 \cdot N_{w,1}$. Because it is possible that $\mathbf{0} \notin S(\mathbf{e})$ and $|S(\mathbf{e})| \geq 2$ (as shown in Fig. 4), the number of errors causing Case 2 decoding failure is at most $254 \cdot N_{w,1}$.

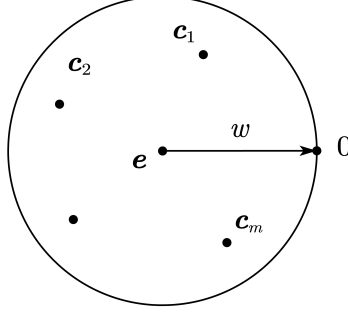


Figure 4: The case where the error vector $\mathbf{e}$ is closer to m nonzero codewords than to the zero codeword.

If $\mathbf{0} \in S(\mathbf{e})$ and $|S(\mathbf{e})| \geq 2$, then $d(\mathbf{c}_1, \mathbf{e}) = d(\mathbf{0}, \mathbf{e}) = w$, which implies $u = 32s$, i.e., $\mathbf{e}$ belongs to Case 3. For a single codeword $\mathbf{c}_1$, the number of error vectors $\mathbf{e}$ of weight $w \leq 64s - 1$ satisfying $u = 32s$ is

$$M_{w,1} = \binom{64s}{32s} \binom{64s}{w-32s}.$$

Similarly, we can conclude that at most $254 \cdot M_{w,1}$ distinct $\mathbf{e}$ satisfy $\mathbf{0} \in S(\mathbf{e})$ and $|S(\mathbf{e})| \geq 2$. Among these error vectors $\mathbf{e}$, it is possible that $|S(\mathbf{e})| = m \geq 3$, as shown in Fig. 5. In this case, according to Eq. (15), the contribution of $S(\mathbf{e})$ to the decoding error count is $\frac{m-1}{m}$. Conversely, if we assume that $m - 1$ $\mathbf{e}$ belonging to Case 3 satisfy $|S(\mathbf{e})| = 2$, then their contribution to the decoding error count is $\frac{m-1}{2}$. Since $m - 1 \geq 2$, we have $\frac{m-1}{2} \geq 1 > \frac{m-1}{m}$, meaning the latter estimation method would overestimate the decoding error count. Therefore, the expected number of errors causing Case 3 decoding failure is at most $254 \cdot \frac{M_{w,1}}{2}$.

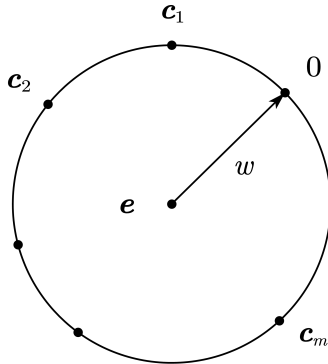


Figure 5: The case where m nonzero codewords are equidistant from the error vector $\mathbf{e}$.

Combining both cases, for $w \leq 64s - 1$, we can estimate that the expected number of errors causing decoding failure does not exceed $E_{w,1} = 254 \cdot N_{w,1} + 254 \cdot \frac{M_{w,1}}{2}$.

As mentioned earlier, this estimation method, which only considers the relationship between a single codeword and the error, tends to overestimate the number of decoding failure errors $E_{w,1}$. To improve accuracy, we also need to consider the intersection relationships between the supports of multiple nonzero codewords and the support of the error.

Considering the relationship between two nonzero codewords $\mathbf{c}_1, \mathbf{c}_2$ and the error vector $\mathbf{e}$: There are $\binom{254}{2}$ unordered pairs of codewords $(\mathbf{c}_1, \mathbf{c}_2)$ of weight $64s$ in the 8-dimensional RM code C , and $\text{wt}(\mathbf{c}_1 + \mathbf{c}_2) = 64s$ or $128s$. If $\text{wt}(\mathbf{c}_1 + \mathbf{c}_2) = 128s$, then $|\text{Supp}(\mathbf{c}_1) \cup \text{Supp}(\mathbf{c}_2)| = 128s$. By Lemma A.1, there is no error vector $\mathbf{e}$ of weight less than $64s$ satisfying $d(\mathbf{c}_1, \mathbf{e}) = d(\mathbf{c}_2, \mathbf{e}) = \text{wt}(\mathbf{e})$ or $d(\mathbf{c}_1, \mathbf{e}), d(\mathbf{c}_2, \mathbf{e}) < \text{wt}(\mathbf{e})$. Therefore, we only need to consider the case $\text{wt}(\mathbf{c}_1 + \mathbf{c}_2) = 64s$. There are $\binom{254}{2} - 127 = 32004$ such codeword pairs.

After appropriate transformation, the matrix formed by codewords $\mathbf{c}_1, \mathbf{c}_2$ and the error vector $\mathbf{e}$ can be reduced to the following form:

$$\begin{pmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{32s} & \mathbf{1}_{32s} & \mathbf{0}_{32s} & \mathbf{0}_{32s} \\ \mathbf{0}_{32s} & \mathbf{1}_{32s} & \mathbf{1}_{32s} & \mathbf{0}_{32s} \\ \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 & \mathbf{e}_4 \end{pmatrix}, \quad (17)$$

where $\text{wt}(\mathbf{e}_i) = u_i$. The distances between codewords $\mathbf{c}_1, \mathbf{c}_2$ and the error vector $\mathbf{e}$ can be expressed as

$$\begin{cases} d(\mathbf{c}_1, \mathbf{e}) = \text{wt}(\mathbf{c}_1) + \text{wt}(\mathbf{e}) - 2|\text{Supp}(\mathbf{c}_1) \cap \text{Supp}(\mathbf{e})| = 64s + w - 2(u_1 + u_2), \\ d(\mathbf{c}_2, \mathbf{e}) = \text{wt}(\mathbf{c}_2) + \text{wt}(\mathbf{e}) - 2|\text{Supp}(\mathbf{c}_2) \cap \text{Supp}(\mathbf{e})| = 64s + w - 2(u_2 + u_3). \end{cases}$$

If $\mathbf{0} \notin S(\mathbf{e})$ and $|S(\mathbf{e})| \geq 2$, then $d(\mathbf{c}_1, \mathbf{e}), d(\mathbf{c}_2, \mathbf{e}) < \text{wt}(\mathbf{e})$, and u_i satisfy $u_1 + u_2 > 32s$ and $u_2 + u_3 > 32s$. In this case, codewords $\mathbf{c}_1, \mathbf{c}_2 \in S(\mathbf{e})$, and the error vector $\mathbf{e}$ causes decoding failure of Case 2. For a single codeword pair $(\mathbf{c}_1, \mathbf{c}_2)$, the total number of errors satisfying this condition is:

$$N_{w,2} = \sum_{u_1=1}^{32s} \binom{32s}{u_1} \sum_{u_2=32s+1-u_1}^{32s} \binom{32s}{u_2} \sum_{u_3=32s+1-u_2}^{32s} \binom{32s}{u_3} \binom{32s}{w-u_1-u_2-u_3}. \quad (18)$$

If $\mathbf{0} \in S(\mathbf{e})$ and $|S(\mathbf{e})| \geq 3$, then $d(\mathbf{c}_1, \mathbf{e}) = d(\mathbf{c}_2, \mathbf{e}) = \text{wt}(\mathbf{e})$, and u_i satisfy $u_1 + u_2 = 32s$ and $u_1 = u_3$. In this case, codewords $\mathbf{0}, \mathbf{c}_1, \mathbf{c}_2 \in S(\mathbf{e})$, and the error vector $\mathbf{e}$ causes decoding failure of Case 3. For a single codeword pair $(\mathbf{c}_1, \mathbf{c}_2)$, the total number of errors satisfying this condition is:

$$M_{w,2} = \sum_{u_1=0}^{32s} \binom{32s}{u_1}^2 \binom{32s}{32s-u_1} \binom{32s}{w-32s-u_1}. \quad (19)$$

By Lemma A.2, when $\text{wt}(\mathbf{e}) < 40s$, the cases with $m \geq 3$ described in Fig. 4 and 5 do not occur. Therefore, when $\text{wt}(\mathbf{e}) < 40s$, all $S(\mathbf{e})$ that do not contain the zero codeword and have $|S(\mathbf{e})| \geq 2$ satisfy $|S(\mathbf{e})| = 2$; all $S(\mathbf{e})$ that contain the zero codeword and have $|S(\mathbf{e})| \geq 3$ satisfy $|S(\mathbf{e})| = 3$. Hence, the expected number of decoding failures can be improved to

$$\begin{aligned} E_{w,2}^1 &= (254 \cdot N_{w,1} - 32004 \cdot N_{w,2}) + \left(\frac{254 \cdot M_{w,1} - 2 \cdot 32004 \cdot M_{w,2}}{2} + \frac{2 \cdot 32004 \cdot M_{w,2}}{3} \right) \\ &= 254 \cdot N_{w,1} + 127 \cdot M_{w,1} - 32004 \cdot N_{w,2} - 10668 \cdot M_{w,2}. \end{aligned} \quad (20)$$

Because when $w \geq 40s$, error vectors $\mathbf{e}$ causing Case 3 decoding failure may satisfy $|S(\mathbf{e})| \geq 4$. Using $127 \cdot M_{w,1} - 10668 \cdot M_{w,2}$ as the estimate for the number of Case 3 decoding failure errors could lead to underestimation. Therefore, for errors with $40s \leq w \leq 40s + 1$, the count is

$$E_{w,2}^2 = 254 \cdot N_{w,1} + 127 \cdot M_{w,1} - 32004 \cdot N_{w,2}. \quad (21)$$

When $w \geq 40s + 2$, $E_{w,1}$ remains the count for the number of decoding failure errors.

Since this counting method is extremely conservative, as w increases, the actual expected number of errors may be far lower than the above estimate $E_{w,1}$, and even $E_{w,1} > \binom{128s}{w}$ may

occur. Therefore, multiple counting problems, as shown in Figs. 4 and 5 must appear. Next, the Cauchy-Schwarz inequality is applied to improve $E_{w,1}$ for the case where $254 \cdot N_{w,1} < \binom{128s}{w}$ and $E_{w,1} > \binom{128s}{w}$.

Improving $E_{w,1}$ using the Cauchy-Schwarz inequality: For $m \geq 1$, let a_m denote the number of distinct errors $\mathbf{e} \in \mathcal{S}_w$ such that $\mathbf{0} \in S(\mathbf{e})$ and $|S(\mathbf{e})| = m$. For such an error $\mathbf{e}$, the probability of successful decoding is $1/m$. When $m = 1$, $\mathbf{e}$ belongs to Case 1; when $m \geq 2$, it belongs to Case 3.

Let the actual number of weight- w errors causing Case 2 decoding failure be $\underline{N}_w$. Since $254 \cdot N_{w,1}$ may overestimate the number of errors causing Case 2 decoding failure, we have $\underline{N}_w \leq 254 \cdot N_{w,1}$. Let the maximum value of m be ν . Then we have

$$\begin{cases} \sum_{m=2}^{\nu} (m-1)a_m = 254 \cdot M_{w,1}, \\ \sum_{m=1}^{\nu} a_m = \binom{128s}{w} - \underline{N}_w \geq \binom{128s}{w} - 254 \cdot N_{w,1}. \end{cases} \quad (22)$$

It follows that

$$\sum_{m=1}^{\nu} ma_m = \binom{128s}{w} - \underline{N}_w + 254 \cdot M_{w,1} \geq \binom{128s}{w} - 254 \cdot N_{w,1} + 254 \cdot M_{w,1}.$$

By the Cauchy-Schwarz inequality,

$$\sum_{m=1}^{\nu} ma_m \cdot \sum_{m=1}^{\nu} \frac{a_m}{m} \geq \left(\sum_{m=1}^{\nu} a_m \right)^2,$$

therefore, the expected number of weight- w errors that are successfully decoded satisfies

$$\sum_{m=1}^{\nu} \frac{a_m}{m} \geq \frac{\left(\binom{128s}{w} - \underline{N}_w \right)^2}{\binom{128s}{w} - \underline{N}_w + 254 \cdot M_{w,1}}.$$

Let $z = \binom{128s}{w} - \underline{N}_w$. Since the function $\frac{z^2}{z + 254 \cdot M_{w,1}}$ is monotonically increasing for $z > 0$, and

$$\binom{128s}{w} - \underline{N}_w \geq \binom{128s}{w} - 254 \cdot N_{w,1},$$

we have

$$\sum_{m=1}^{\nu} \frac{a_m}{m} \geq \frac{\left(\binom{128s}{w} - 254 \cdot N_{w,1} \right)^2}{\binom{128s}{w} - 254 \cdot N_{w,1} + 254 \cdot M_{w,1}}.$$

Therefore, in this case, $E_{w,1}$ can be improved to

$$E'_{w,1} = \binom{128s}{w} - \frac{\left(\binom{128s}{w} - 254 \cdot N_{w,1} \right)^2}{\binom{128s}{w} - 254 \cdot N_{w,1} + 254 \cdot M_{w,1}}.$$

Combining all the above counting cases, we obtain the failure probability profile for the RM code. $\square$

A-B.2 Proof of Theorem V.6

Proof of Theorem V.6. Since the X_b are independent and each X_b follows a Binomial distribution $\text{Binomial}(\zeta_b, p_{\text{rm}}(b))$, its probability generating function is

$$G_{X_b}(z) = (1 - p_{\text{rm}}(b) + p_{\text{rm}}(b)z)^{\zeta_b}.$$

By independence, the probability generating function of e_{rm} is the product of the probability generating functions of the individual X_b :

$$G_{e_{rm}}(z) = \prod_{b=0}^m (1 - p_{rm}(b) + p_{rm}(b)z)^{\zeta_b}.$$

Let $G_{e_{rm}}(z) = \sum_{j \geq 0} a_j z^j$, where $a_k = \text{coeff}(G_{e_{rm}}(z), z^k)$. By the properties of probability generating functions, we have $\Pr(e_{rm} = k) = a_k$. Therefore,

$$\sum_{j=0}^t \Pr(e_{rm} = j) = \sum_{j=0}^t a_j = \sum_{j=0}^t \text{coeff} \left(\prod_{b=0}^m (1 - p_{rm}(b) + p_{rm}(b)z)^{\zeta_b}, z^j \right).$$

Thus,

$$\Pr(e_{rm} > t) = 1 - \Pr(e_{rm} \leq t) = 1 - \sum_{j=0}^t \text{coeff} \left(\prod_{b=0}^m (1 - p_{rm}(b) + p_{rm}(b)z)^{\zeta_b}, z^j \right).$$

□

A-C Proofs for BSC and Weight-Redistribution Results

Proof of Theorem V.7. By cyclic symmetry and the convolution operation, the probability $p(\delta) = \Pr[e'_k = 1]$ does not depend on the index k . The expected Hamming weight is therefore

$$\mathbb{E}[\text{wt}(\mathbf{e}')] = n \cdot p(\delta).$$

Now increase δ by 1. The corresponding weight changes are:

- $w_{\mathbf{e}}$ increases by 2;
- $w_{\mathbf{r}_1}$ and $w_{\mathbf{r}_2}$ each decrease by 1.

Let $p = p(\delta)$ denote the current error probability. The two effects are:

Effect of increasing $w_{\mathbf{e}}$ by 2: Adding two random non-zero entries to $\mathbf{e}$ is equivalent to independently flipping two uniformly random coordinates. For each coordinate flip, if the coordinate was 0, it becomes 1 (increasing weight by 1); if it was 1, it becomes 0 (decreasing weight by 1). The expected change per flip is $(1-p) \cdot 1 + p \cdot (-1) = 1-2p$. For two independent flips:

$$\Delta_{\mathbf{e}} = 2(1-2p).$$

Effect of decreasing $w_{\mathbf{r}_1}$ and $w_{\mathbf{r}_2}$ by 1: Decreasing $w_{\mathbf{r}_1}$ by 1 means removing one random 1-entry, which is equivalent to $\mathbf{e}'$ with a random cyclic shift of $\mathbf{y}$. Since $\mathbf{y}$ has weight $w_{\mathbf{x}}$ (note that $w_{\mathbf{x}} = w_{\mathbf{y}}$), this flips exactly $w_{\mathbf{x}}$ coordinates. Similarly, decreasing $w_{\mathbf{r}_2}$ flips another $w_{\mathbf{x}}$ coordinates.

If the two sets of flipped positions were entirely disjoint, the expected weight change would be $2w_{\mathbf{x}}(1-2p)$. However, two independent random cyclic shifts overlap in expectation at $\mathbb{E}[\text{overlap}] = w_{\mathbf{x}}^2/n$ positions. At each overlapping position, both flips cancel each other, contributing 0 to the net weight change. Therefore, the net expected change is:

$$\Delta_{\mathbf{r}} = 2w_{\mathbf{x}}(1-2p) - 2 \cdot \frac{w_{\mathbf{x}}^2}{n}(1-2p) = 2(1-2p) \left(w_{\mathbf{x}} - \frac{w_{\mathbf{x}}^2}{n} \right).$$

Total change: The net change in expected weight when $\delta \rightarrow \delta + 1$ is

$$\Delta = \Delta_{\mathbf{e}} - \Delta_{\mathbf{r}} = 2(1-2p) - 2(1-2p) \left(w_{\mathbf{x}} - \frac{w_{\mathbf{x}}^2}{n} \right) = 2(1-2p) \left(1 - w_{\mathbf{x}} + \frac{w_{\mathbf{x}}^2}{n} \right).$$

By the theorem's hypotheses, $w_{\mathbf{x}} \geq 67$ and $p < 0.5$. We have $1 - 2p > 0$ and

$$1 - w_{\mathbf{x}} + \frac{w_{\mathbf{x}}^2}{n} < 1 - 67 + \frac{w_{\mathbf{x}}^2}{n} < 0.$$

Therefore, $\Delta < 0$ for all $0 \leq \delta \leq w_{\mathbf{r}} - w_{\mathbf{x}}$. Since $\mathbb{E}[\text{wt}(\mathbf{e}')] = n \cdot p(\delta)$ strictly decreases with δ , the marginal error probability $p(\delta)$ itself is strictly decreasing in δ . $\square$

Proof of Theorem V.8. For fixed positive integers A and B , first consider $\tilde{p}(n; A, B)$ as a function of n . From Lemma V.2, for a fixed odd ℓ , the term

$$T_{\ell}(n) = \frac{\binom{A}{\ell} \binom{n-A}{B-\ell}}{\binom{n}{B}}$$

can be rewritten as

$$T_{\ell}(n) = \binom{A}{\ell} \cdot \frac{(n-A)(n-A-1) \cdots (n-A-(B-\ell)+1)}{n(n-1) \cdots (n-B+1)} \cdot \frac{B!}{(B-\ell)!}.$$

Consider the ratio

$$\frac{T_{\ell}(n+1)}{T_{\ell}(n)} = \frac{(n+1-A)(n+1-B)}{(n+1)(n+1-A-B+\ell)}.$$

Since $n \geq A+B$ and $\ell \geq 1$, we have

$$(n+1-A)(n+1-B) < (n+1)(n+1-A-B+\ell) \iff AB < (n+1)\ell.$$

For fixed A, B , when n is sufficiently large (specifically, $n > AB - 1$), we obtain $AB < n+1 \leq (n+1)\ell$, hence $T_{\ell}(n+1) < T_{\ell}(n)$. Because each nonzero term $T_{\ell}(n)$ is positive and strictly decreasing, their sum $\tilde{p}(n; A, B)$ is also strictly decreasing.

Therefore, $p_U(n)$, $p_V(n)$, and $p_W(n)$ are all strictly decreasing in n . Now, from Lemma V.3, computing the partial derivatives yields

$$\frac{\partial P}{\partial p_U} = (1 - 2p_V)(1 - 2p_W), \quad \frac{\partial P}{\partial p_V} = (1 - 2p_U)(1 - 2p_W), \quad \frac{\partial P}{\partial p_W} = (1 - 2p_U)(1 - 2p_V).$$

Under typical parameter choices, $p_U, p_V, p_W < 1/2$, so all partial derivatives are positive. Thus P is strictly increasing in each of p_U, p_V, p_W . Since $p_U(n)$, $p_V(n)$, and $p_W(n)$ are all strictly decreasing in n and P is strictly increasing in each of them, the composite function $P(n)$ is strictly decreasing in n . $\square$

Proof of Theorem B.1. Treat k as a fixed integer parameter, with $k \rightarrow \infty$. Since the denominator $\prod \binom{k}{w_i}$ is independent of the decision variables $(k_i)_{i=1}^s$, maximizing P_{struct} is equivalent to maximizing

$$\Phi(k_1, \dots, k_s) := \prod_{i=1}^s \binom{k - k_i}{w_i} \quad \text{subject to} \quad \sum_{i=1}^s k_i = k. \quad (23)$$

1. Balanced Case ($\delta = 0$). Here $w_1 = \dots = w_s = \bar{w}$. The function $f(x) = \binom{k-x}{w}$ is log-concave [15]. Specifically, the ratio of adjacent terms is

$$\rho_w(x) := \frac{\binom{k-(x+1)}{w}}{\binom{k-x}{w}} = \frac{k-x-w}{k-x} = 1 - \frac{w}{k-x}.$$

For $w \geq 0$, the function $x \mapsto 1 - \frac{w}{k-x}$ is strictly decreasing in x . Due to the symmetry of the weights and the strict log-concavity, the product is maximized when the arguments k_i are balanced. Thus, asymptotically $k_1 = \dots = k_s = k/s$.

2. Unbalanced Case ($\delta > 0$). In this case, $w_1 = \dots = w_{s-1} < w_s$. Applying the balancing argument from Step 1 to the indices $1, \dots, s-1$ (which share the identical weight w_1) implies that any maximizer must asymptotically satisfy

$$k_1 = \dots = k_{s-1}.$$

Let $k_1 = k_{s-1} = a$. The constraint $\sum k_i = k$ implies $k_s = k - (s-1)a$.

It remains to show that $a > k_s$. Consider a unit transfer from k_s to k_1 , i.e., $(a, k_s) \rightarrow (a+1, k_s-1)$. The ratio of the new objective value to the old is:

$$R(a, k_s) := \frac{\binom{k-a-1}{w_1}}{\binom{k-a}{w_1}} \cdot \frac{\binom{k-k_s+1}{w_s}}{\binom{k-k_s}{w_s}} = \rho_{w_1}(a) \cdot \frac{1}{\rho_{w_s}(k_s-1)}. \quad (24)$$

Substituting $\rho_w(x) = 1 - \frac{w}{k-x}$, we obtain:

$$R(a, k_s) = \frac{1 - \frac{w_1}{k-a}}{1 - \frac{w_s}{k-k_s+1}}.$$

Assume for the sake of contradiction that $a \leq k_s$. Since $w_1 < w_s$ and $k-a \geq k-k_s$, it follows that:

$$\frac{w_1}{k-a} \leq \frac{w_1}{k-k_s} < \frac{w_s}{k-k_s+1} \implies 1 - \frac{w_1}{k-a} > 1 - \frac{w_s}{k-k_s+1}.$$

This implies $R(a, k_s) > 1$, meaning that transferring mass from k_s to k_1 increases the objective function. This contradicts the optimality of the allocation if $a \leq k_s$. Therefore, the maximizer must satisfy $a > k_s$, i.e., $k_1 > k_s$.

3. Monotonicity with respect to δ . We employ a continuous relaxation of the problem. Let k_i take real values and replace factorials with Γ functions. Define the optimal value function:

$$V(\delta) := \max_{\mathbf{k} \in \mathbb{R}^s, \sum k_i = k} \ln P_{\text{struct}}(\mathbf{k}; \delta).$$

By the Envelope Theorem, the derivative $V'(\delta)$ is given by the partial derivative of the Lagrangian with respect to δ , evaluated at the optimal solution $\mathbf{k}^*$. Recall that $\frac{dw_1}{d\delta} = -1$ and $\frac{dw_s}{d\delta} = s-1$. Thus:

$$V'(\delta) = (s-1) \left[\frac{\partial}{\partial w_s} \ln \frac{\binom{k-k_s^*}{w_s}}{\binom{k}{w_s}} - \frac{\partial}{\partial w_1} \ln \frac{\binom{k-k_1^*}{w_1}}{\binom{k}{w_1}} \right].$$

Using the identity $\partial_w \ln \binom{M}{w} = \psi(M-w+1) - \psi(w+1)$, the derivative of the term $\ln \frac{\binom{M}{w}}{\binom{M}{k}}$ with respect to w is $\psi(M-w+1) - \psi(k-w+1)$. Let $m_i = k - k_i^*$. The expression for $V'(\delta)$ simplifies to:

$$\frac{V'(\delta)}{(s-1)} = ([\psi(m_s - w_s + 1) - \psi(k - w_s + 1)] - [\psi(m_1 - w_1 + 1) - \psi(k - w_1 + 1)]).$$

Using the asymptotic expansion $\psi(x) = \ln x + O(1/x)$ and the fact that $w_i = o(k)$, we have:

$$\psi(m_i - w_i + 1) - \psi(k - w_i + 1) = \ln \left(\frac{m_i - w_i}{k - w_i} \right) + o(1) = \ln \left(\frac{m_i}{k} \right) + o(1).$$

Therefore, asymptotically:

$$V'(\delta) \approx (s-1) \left(\ln \frac{m_s}{k} - \ln \frac{m_1}{k} \right) = (s-1) \ln \left(\frac{k - k_s^*}{k - k_1^*} \right).$$

From Part 2, we established that $k_1^* > k_s^*$, which implies $k - k_s^* > k - k_1^*$. Consequently, the logarithm is strictly positive, yielding $V'(\delta) > 0$. This proves that the maximum value is strictly increasing in δ . $\square$

B Structured-ISD Analysis for Unbalanced QCSD

This section gives the structured-error ISD calculation used in the concrete security evaluation in Section V-C.

Expected Permutation Cost for Structured Errors

Suppose the error vector follows a *structured* distribution: the length n is partitioned into $s \geq 2$ blocks of lengths n_i and each block has fixed weight w_i (with $\sum_i n_i = n$ and $\sum_i w_i = w$). Then the error space shrinks from $\binom{n}{w}$ to $\prod_{i=1}^s \binom{n_i}{w_i}$. Accordingly, the classical expression P_{generic} no longer applies.

Let k_i denote how many coordinates of block i fall into the chosen $(k+l)$ -bit information set, and let p_i denote the number of errors from block i inside that information set, where $\sum_i k_i = k+l$ and $\sum_i p_i = p$. Conditioned on the block weights $(w_i)_{i=1}^s$, the number of consistent splittings is $\prod_{i=1}^s \binom{k_i}{p_i} \binom{n_i - k_i}{w_i - p_i}$, and dividing by the structured error-space size yields the permutation success probability. For a given choice of (l, p) and the (effective) allocation (k_i, p_i) , the success probability is

$$P_{\text{struct}} = \prod_{i=1}^s \frac{\binom{k_i}{p_i} \binom{n_i - k_i}{w_i - p_i}}{\binom{n_i}{w_i}}, \quad (25)$$

and the corresponding expected permutation cost is $T'_p = P_{\text{struct}}^{-1}$.

The asymptotic discussion uses the standard HQC scaling $w = o(n)$. Under this regime, [16] gives the usual optimal behavior $l = o(n)$ and $p = o(w)$ as $n \rightarrow \infty$. We therefore examine the maximizing allocation for P_{struct} in the limiting regime $p = l = 0$, and study how that maximum changes under imbalance. For HQC, we have $s = 2, 3$, $n_i = k = n/s$, and $P_{\text{struct}} = \prod_{i=1}^s \frac{\binom{k - k_i}{w_i}}{\binom{k}{w_i}}$.

Theorem B.1. *Let $s \in \{2, 3\}$. Assume $k \rightarrow \infty$, $w = o(k)$, and denote $\bar{w} = w/s$. Fix an integer δ such that $0 \leq \delta \leq \bar{w}$. Define the block weights as*

$$w_1 = \dots = w_{s-1} = \bar{w} - \delta, \quad w_s = \bar{w} + (s-1)\delta,$$

satisfying $\sum_{i=1}^s w_i = w$. Consider the optimization of $P_{\text{struct}} = \prod_{i=1}^s \frac{\binom{k - k_i}{w_i}}{\binom{k}{w_i}}$ subject to $\sum_{i=1}^s k_i = k$ and $0 \leq k_i \leq k - w_i$. We have the following results:

1. *If $\delta = 0$, asymptotically any maximizer $(k_i)_{i=1}^s$ satisfies $k_1 = \dots = k_s$.*
2. *If $\delta > 0$, asymptotically any maximizer $(k_i)_{i=1}^s$ satisfies $k_1 = k_{s-1} > k_s$.*
3. *The maximum value of P_{struct} is strictly increasing with respect to δ .*

Proof location. See Appendix A-C.

Remark B.2. Theorem B.1 is asymptotic. For finite parameter sets, integrality constraints matter, and the $(k_i)_{i=1}^s$ typically fall into one of two regimes: either

$$k_1 \approx \dots \approx k_s \quad \text{or} \quad k_1 \approx k_{s-1} \gtrsim k_s.$$

In the specific ISD security calculation, the theorem restricts the candidate allocations (k_i, p_i) that need to be evaluated.

Finite-parameter check. Set $w_{\mathbf{r}_1} = w_{\mathbf{r}_2} = w_{\mathbf{r}} - \delta$ and $w_{\mathbf{e}} = w_{\mathbf{r}} + 2\delta$. Fig. 6 shows the corresponding permutation bit-complexity $\log_2 T'_p = \min_{k_i, p_i} \{\log_2 P_{\text{struct}}^{-1}\}$ for HQC parameters. The curve decreases monotonically with δ . This illustrates the trade-off used in the TriQ-KEM weight choice: increasing the imbalance lowers the induced BSC error rate and also lowers the permutation cost considered in the structured-ISD estimate.

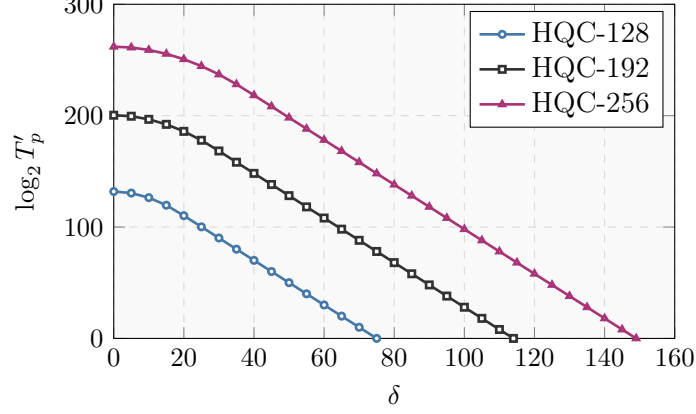


Figure 6: Comparison of the permutation bit-complexity $\log_2 T'_p$ for $3\text{-CW-QCSD}(3n, w_{\mathbf{r}} - \delta, w_{\mathbf{r}} - \delta, w_{\mathbf{r}} + 2\delta)$ problem with $p = 0$ and $l = 0$, where n and $w_{\mathbf{r}}$ are parameters in HQC sets, respectively.

Oracle Complexity under Structured Errors

In the oracle step of concrete ISD algorithms, one searches for vectors $\mathbf{e}_1 \in S_{\mathbf{e}_1}$ satisfying $\mathbf{H}_1 \mathbf{e}_1^\top = \mathbf{s}_1^\top$. For each candidate $\mathbf{e}_1$, the corresponding $\mathbf{e}_2$ is computed as $\mathbf{e}_2^\top = \mathbf{s}_2^\top - \mathbf{H}_2 \mathbf{e}_1^\top$ and accepted if $\text{wt}(\mathbf{e}_2) = w - p$ holds.

Now assume that $\mathbf{e}_1$ is *structured*: the $(k + l)$ coordinates are partitioned into $s \geq 2$ disjoint blocks of sizes $(k_1, \dots, k_s)$ with $\sum_{i=1}^s k_i = k + l$, and the attacker additionally knows that the weight p splits as $(p_1, \dots, p_s)$ with $\sum_{i=1}^s p_i = p$, i.e., $\text{wt}(\mathbf{e}_1^{(i)}) = p_i$ in block i . Then, the space containing structured $\mathbf{e}_1$ can be written as $S'_{\mathbf{e}_1} = \left\{ \mathbf{e}_1 \in \mathbb{F}_2^{k+l} : \text{wt}(\mathbf{e}_1^{(i)}) = p_i \text{ for } i = 1, \dots, s \right\}$.

Let $\mathcal{A}$ be any oracle algorithm for the unstructured setting with complexity T_o , and let $\mathcal{A}'$ be its structured counterpart. Concrete ISD oracles choose a search space $S_{\mathbf{e}_1}$ that is favorable for success. For example, Stern's meet-in-the-middle step works best when $\mathbf{e}_1$ is split into two equal halves [8]. Once the attacker is restricted to the structured space $S'_{\mathbf{e}_1}$, this advantage does not improve. For a fixed $\text{SMD}(k + l, k, p)$ instance, the success probability of $\mathcal{A}'$ is therefore no larger than that of $\mathcal{A}$. Hence $\mathcal{A}'$ runs on average in time

$$T_{\mathcal{A}'} \geq \frac{|S'_{\mathbf{e}_1}|}{|S_{\mathbf{e}_1}|} \cdot T_{\mathcal{A}} = \frac{\prod_i \binom{k_i}{p_i}}{|S_{\mathbf{e}_1}|} \cdot T_{\mathcal{A}}.$$

Thus, we have $T'_o \geq T_o/2^I$, where $I = \max \left\{ \log_2 |S_{\mathbf{e}_1}| - \sum_{i=1}^s \log_2 \binom{k_i}{p_i}, 0 \right\}$. The overall ISD bit-complexity under structured errors is therefore conservatively (attacker-favorable) estimated by

$$T'_{\text{ISD}} = \min_{l, p, \{k_i, p_i\}} \left(\log_2 T'_p + \log_2 (T_{\text{gauss}} + T_o/2^I) \right). \quad (26)$$

For convenience, given an error vector $\mathbf{e}$ with known weight distribution, we denote by $T_{\text{ISD}}(n, k, \mathbf{e})$ the ISD bit-complexity of the corresponding $s\text{-CW-QCSD}$ problem under the DOOM attack [12].¹ Following the calculation above, the ISD bit-complexity for a structured error $\mathbf{e}$ satisfies

$$\underline{T}_{\text{ISD}}(n, k, \mathbf{e}) = \min \{ T_{\text{ISD}}, T'_{\text{ISD}} \} - \log_2 \sqrt{\#\text{targets}}. \quad (27)$$

¹For the $s\text{-CW-QCSD}(sn, w_1, w_2, \dots, w_s)$ problem, we have $\#\text{targets} = n$.

C HQC Parameters and Estimates for Comparison

This appendix section gives the HQC parameter sets, cycle counts, ISD estimates, and CDFR estimates used for comparison. The standardized HQC instances are cited from the HQC specification, while the 384-bit and 512-bit entries are auxiliary extrapolated reference points used only for comparison in this document.

C-A HQC Parameter Sets Used for Comparison

This section gives the standardized HQC parameter sets [1] together with the auxiliary HQC-384 and HQC-512 sets used for comparison in this specification. The standardized HQC instances (HQC-128, HQC-192, HQC-256) are taken from the HQC specification. The auxiliary HQC-384 and HQC-512 instances are extrapolated from the same design methodology: the ring length n is chosen as the smallest primitive prime exceeding the RSRM codeword length $n_1 n_2$; the RM block length n_2 grows with the security target; the outer RS code parameters (n_1, k_{RS}) are selected to keep the estimated natural DFR below $2^{-\lambda}$; and the weights are balanced, i.e., $w_{\mathbf{x}} = w_{\mathbf{y}}$ and $w_{\mathbf{r}} = w_{\mathbf{e}}$, where $w_{\mathbf{r}}$ is the common weight of $\mathbf{r}_1$ and $\mathbf{r}_2$, with $w_{\mathbf{x}}/w_{\mathbf{r}} \approx 0.88$ as in the standard HQC sets. The HQC-384 and HQC-512 instances are not part of the NIST HQC submission; they are auxiliary reference points used in this document only for the size and runtime comparisons at the 384-bit and 512-bit classical comparison targets.

Table 15: Standard and extrapolated HQC parameter sets used for comparison

Instance	n_1	n_2	$n_1 n_2$	n	k_{RS}	w	$w_{\mathbf{r}} = w_{\mathbf{e}}$	Natural DFR	Source
HQC-128	46	384	17664	17669	16	66	75	$\leq 2^{-132.89}$	[1]
HQC-192	56	640	35840	35851	24	100	114	$\leq 2^{-193.86}$	[1]
HQC-256	90	640	57600	57637	32	131	149	$\leq 2^{-260.60}$	[1]
HQC-384	130	896	116480	116483	48	196	223	$\leq 2^{-388.31}$	extrapolated
HQC-512	166	1152	191232	191237	64	260	295	$\leq 2^{-514.18}$	extrapolated

All listed HQC instances use balanced fixed weights for the PKE encryption vectors, namely $w_{\mathbf{r}} = w_{\mathbf{e}}$ with $w_{\mathbf{r}}$ common to $\mathbf{r}_1$ and $\mathbf{r}_2$, with the natural DFR estimated via the HQC BSC model [1] and $w/w_{\mathbf{r}} \approx 0.88$. In the API_PKC HQC code used for comparison, the seed length and shared-secret length are 16, 32, 48, and 64 bytes for the 128-, 256-, 384-, and 512-bit comparison levels, respectively; the ciphertext salt length is 16 bytes for all listed HQC instances.

Table 16 records the measured HQC reference cycle counts used in the runtime comparison of Section VI. The benchmarking conditions match those described in that section: MSYS2 UCRT64 with GCC 15.2.0, `-std=c99 -O3`, RDTSC cycle measurements, and 10 000 repetitions per operation. The reported values are per-call medians.

Table 16: HQC reference cycle counts used for comparison (median cycles per call, 10 000 repetitions)

Instance	KeyGen	Encaps	Decaps	Total
HQC-128	4 367 571	8 419 044	12 731 609	25 518 224
HQC-256	27 303 132	54 149 813	81 624 237	163 077 182
HQC-384	84 072 416	167 482 187	252 261 030	503 815 633
HQC-512	151 314 232	299 869 152	450 564 790	901 748 174

The HQC reference cycle counts correspond to an HQC reference implementation whose polynomial multiplication uses a Toom–Cook 3-way decomposition followed by recursive Karatsuba [1]. The reference path operates on scalar `uint64_t` words. The internal working sizes

n_{mult} for the Toom–Cook decomposition are as follows: 18,048 bits for HQC-128, 36,480 bits for HQC-192, 59,904 bits for HQC-256, 117,504 bits for HQC-384, and 194,304 bits for HQC-512. These correspond respectively to 282, 570, 936, 1,836, and 3,036 scalar words.

C-B HQC ISD Estimates for Comparison

Table 17 reports the ISD bit-complexity of the 2-CW-QCSD instance $\underline{T}_{\text{ISD}}(2n, n, (\mathbf{x}, \mathbf{y}))$ for each listed HQC parameter set. Table 18 reports the corresponding complexity for the 3-CW-QCSD instance $\underline{T}_{\text{ISD}}(3n, n, (\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}))$. The estimation methodology follows Eqs. (13) and (12) in Section V-C and is evaluated using the same ISD estimator as the TriQ-KEM sets.

Table 17: ISD bit-complexity for the HQC 2-CW-QCSD instance $\underline{T}_{\text{ISD}}(2n, n, (\mathbf{x}, \mathbf{y}))$

Parameter set	Prange	Stern	BothMay	MayOzerov	BJMM	Q-Prange
HQC-128	166.33	145.31	145.44	145.44	145.63	94.39
HQC-192	236.80	213.45	213.58	213.58	213.71	130.46
HQC-256	300.46	275.60	275.76	275.76	275.82	162.84
HQC-384	432.93	405.81	405.80	405.69	405.99	229.90
HQC-512	562.68	533.97	533.57	533.57	533.80	295.34

Table 18: ISD bit-complexity for the HQC 3-CW-QCSD instance $\underline{T}_{\text{ISD}}(3n, n, (\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}))$

Parameter set	Prange	Stern	BothMay	MayOzerov	BJMM	Q-Prange
HQC-128	167.98	147.25	147.33	147.33	147.52	96.70
HQC-192	238.90	215.79	215.88	215.88	216.02	132.98
HQC-256	301.99	277.36	277.46	277.46	277.55	165.08
HQC-384	434.33	407.41	407.49	407.37	407.56	232.06
HQC-512	562.43	533.92	533.65	533.65	533.85	296.68

For all HQC parameter sets, the minimum classical ISD estimate is determined by the 2-CW-QCSD instance (bold entries in Table 17). The HQC-128 and HQC-256 rows are reproduced in the main ISD comparison tables; the remaining HQC rows are kept as appendix reference data.

C-C HQC DFR and CDFR Estimates for Comparison

Table 19 records the attack-aware DFR and cumulative decryption-failure probability (CDFR) for the HQC parameter sets under the density-interval attack [2], evaluated by the same PEB-BSC methodology as the TriQ-KEM sets in Section V. The classical ISD budgets are taken as the minimum of the 2-CW-QCSD and 3-CW-QCSD minima in Tables 17 and 18 for each HQC parameter set, matching the same convention used for TriQ-KEM. The natural DFR is the decryption-failure rate without an adversarially crafted dense interval.

Table 19: Maximum CDFR for the HQC parameter sets under the density-interval attack [2] (classical setting, $q_d \leq 2^{80}$)

Parameter set	(l_1, l_0)	$\mathcal{T}_p(l_1, l_0)$	DFR(l_1, l_0)	CDFR	Natural DFR
HQC-128	(23, 12)	62.43	$\leq 2^{-115.19}$	$\leq 2^{-35.19}$	$\leq 2^{-132.89}$
HQC-192	(43, 22)	131.63	$\leq 2^{-149.21}$	$\leq 2^{-69.21}$	$\leq 2^{-193.86}$
HQC-256	(61, 31)	198.86	$\leq 2^{-186.70}$	$\leq 2^{-109.97}$	$\leq 2^{-260.60}$
HQC-384	(93, 47)	326.75	$\leq 2^{-261.03}$	$\leq 2^{-182.08}$	$\leq 2^{-388.31}$
HQC-512	(123, 62)	453.42	$\leq 2^{-327.19}$	$\leq 2^{-247.19}$	$\leq 2^{-514.18}$

References

- [1] P. Gaborit, C. Aguilar-Melchor, N. Aragon, S. Bettaiieb, L. Bidoux, O. Blazy, J.-C. Deneuville, E. Persichetti, G. Zémor, J. Bos *et al.*, “Hamming quasi-cyclic (HQC),” PQC-HQC Team, https://pqc-hqc.org/doc/hqc_specifications_2025_08_22.pdf, Tech. Rep., Aug. 2025.
- [2] Q. Guo and T. Johansson, “A new decryption failure attack against HQC,” in *Advances in Cryptology – ASIACRYPT 2020*, S. Moriai and H. Wang, Eds. Cham: Springer International Publishing, 2020, pp. 353–382.
- [3] E. Fujisaki and T. Okamoto, “Secure integration of asymmetric and symmetric encryption schemes,” in *Advances in Cryptology – CRYPTO’ 99*, M. Wiener, Ed. Berlin, Heidelberg: Springer Berlin Heidelberg, 1999, pp. 537–554.
- [4] D. Hofheinz, K. Hövelmanns, and E. Kiltz, “A modular analysis of the Fujisaki-Okamoto transformation,” in *Theory of Cryptography - 15th International Conference – TCC 2017*, ser. LNCS, Y. Kalai and L. Reyzin, Eds., vol. 10677. Springer, 2017, pp. 341–371.
- [5] R. Avanzi, J. Bos, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, J. M. Schanck, P. Schwabe, G. Seiler, D. Stehlé *et al.*, “Crystals-kyber algorithm specifications and supporting documentation,” *NIST PQC Round*, vol. 2, no. 4, pp. 1–43, 2019.
- [6] C. Guan, L. Luo, H. Jiang, J. Hou, T. Yu, H. Wang, K. Li, and L. Qu, “Compact HQC with new (un)balance,” IACR Cryptology ePrint Archive, Report 2026/461, 2026. [Online]. Available: <https://eprint.iacr.org/2026/461>
- [7] E. Prange, “The use of information sets in decoding cyclic codes,” *IRE Transactions on Information Theory*, vol. 8, no. 5, pp. 5–9, 1962.
- [8] J. Stern, “A method for finding codewords of small weight,” in *International colloquium on coding theory and applications*. Springer, 1988, pp. 106–113.
- [9] L. Both and A. May, “Decoding linear codes with high error rate and its impact for LPN security,” in *International Conference on Post-Quantum Cryptography*. Springer, 2018, pp. 25–46.
- [10] A. May and I. Ozerov, “On computing nearest neighbors with applications to decoding of binary linear codes,” in *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. Springer, 2015, pp. 203–228.
- [11] A. Becker, A. Joux, A. May, and A. Meurer, “Decoding random binary linear codes in $2^{n/20}$: How $1+1=0$ improves information set decoding,” in *Annual international conference on the theory and applications of cryptographic techniques*. Springer, 2012, pp. 520–536.
- [12] N. Sendrier, “Decoding one out of many,” in *International Workshop on Post-Quantum Cryptography*. Springer, 2011, pp. 51–67.
- [13] H. N. Ward, “An introduction to divisible codes,” *Des. Codes Cryptography*, vol. 17, no. 1–3, pp. 73–79, Sep. 1999.
- [14] T. Helleseth, T. Klove, and J. Mykkeltveit, “On the covering radius of binary codes,” *IEEE Transactions on Information Theory*, vol. 24, no. 5, pp. 627–628, 1978.
- [15] W. Ehm, “Binomial approximation to the Poisson binomial distribution,” *Statistics & Probability Letters*, vol. 11, no. 1, pp. 7–16, 1991.

- [16] R. Canto Torres and N. Sendrier, “Analysis of information set decoding for a sub-linear error weight,” in *Post-Quantum Cryptography*, T. Takagi, Ed. Cham: Springer International Publishing, 2016, pp. 144–161.